The policy margin is rarely linear

Emergency tutoring across five developing countries, maternal-care incentives in Nigeria, and a tax-formality classic from Bolivia · three rigorous research articles, including two peer-reviewed economics-journal papers. The following article pages are edited reading excerpts, not complete replacements for the originals.

LITERATURE PULSE

All three papers ask what happens beyond the average treatment effect. Phone tutoring works across five countries, but its largest gains appear where school closures leave the weakest counterfactual and live calls outperform light-touch messages. In Nigeria, a small maternal-care payment captures most of the behavioral response, with sharply diminishing returns to larger transfers. In Bolivia, tax registration helps only a middle range of firms while imposing net costs elsewhere in the size distribution. **Literature trend:** applied economics is moving from 'does it work?' toward mapping response curves, delivery conditions, and complier groups. That is the information policymakers need to choose intensity, target populations, and scale without assuming that a larger intervention or a broader mandate is automatically better.

01 CURRENT / EDUCATION AND RESILIENCE · STARTS ON PAGE 2 · Nature 657 · 2026 · peer reviewed · CC BY 4.0

Mobile Education Builds Resilience during Shocks in Five Countries

Can low-tech tutoring keep children learning when disease, typhoons, or other shocks close schools? Five randomized trials in India, Kenya, Nepal, the Philippines, and Uganda compare SMS nudges, targeted phone tutorials, and usual services, including government- and NGO-delivery arms. Live calls raise numeracy by about 0.32 standard deviations across countries; SMS alone averages 0.08, with mixed site-level results. The authors estimate roughly four years of high-quality instruction per $100 spent. Randomization and harmonized assessments support the causal claim, but effects are largest where the emergency counterfactual was weakest, implementation improved over time, and a reachable phone was required. The paper is unusually useful evidence on whether an intervention survives both cross-country replication and government delivery.

02 CURRENT / GLOBAL HEALTH AND INCENTIVES · STARTS ON PAGE 12 · Journal of Development Economics 183 · 2026 · peer reviewed · CC BY-NC 4.0

Right-sizing Incentives

How much cash is needed to move expectant mothers into formal care? In a randomized trial covering 36,607 women in Nigeria, the authors compare conditional payments of N2,500, N4,000, and N10,000 plus a separate pregnancy-risk information treatment. The smallest payment nearly doubles facility delivery—from 28% to about 55%—while larger amounts add little. Incentives reduce stillbirths by about 1.2 percentage points from a 5% baseline and increase later healthcare use for index children and siblings; information alone has little effect. Covid-era implementation changes complicate pooling, and the mortality mechanism is not fully pinned down. The striking result is the response curve: most of the health gain arrives at the extensive margin, so program design depends more on finding the participation threshold than on maximizing the transfer.

03 CLASSIC / INFORMALITY AND FIRM PRODUCTIVITY · STARTS ON PAGE 19 · Journal of Development Economics 91(1) · 2010 · peer reviewed classic · CC BY 3.0 IGO edition

Does It Pay Firms to Register for Taxes? The Impact of Formality on Firm Profitability

Does formal tax registration make small firms more profitable? Using a survey of Bolivian microenterprises and distance to the tax office—conditional on distance to the city center—as an instrument, McKenzie and Sakho estimate a local effect for firms whose registration responds to access. Registration raises profits for firms in the middle of the size distribution, largely through more customers and sales, but is associated with lower profits for the smallest and largest firms. The instrument's exclusion restriction, monotonicity, survey-based profit measurement, and the difference between local and average treatment effects all matter. This classic is a useful warning against a single 'formality premium': the same legal status can relieve a market-access constraint for one firm and impose a tax or compliance cost on another.

Editorial note: references and appendices were omitted where possible; no included article contributes more than about 15 text pages, excluding table-heavy pages. Full citations and source links remain in the catalog. AI-written synopses were checked against the included source pages.

Article

Mobile education builds resilience during shocks in five countries

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Education systems need to withstand frequent shocks, including disease and climate events, which close schools for more than 222 million children[1]. During these emergencies, alternative models are needed to deliver education. However, rigorous evaluation of effective educational approaches in these settings is challenging and rare, especially across multiple countries. Here we present results from five randomized trials in India, Kenya, Nepal, the Philippines and Uganda to evaluate the provision of education in emergency settings. We test multiple scalable models of remote targeted tutoring instruction, comparing delivery by governments and non-governmental organizations (NGOs). Whereas text messages have mixed results, tutorials conducted via phone call show effectiveness across diverse settings. We find large and robust effect sizes on learning. These effects are highly cost-effective, and deliver approximately 4 years of high-quality instruction per US$100 spent. Results show large effects when delivered by government teachers or NGO instructors. Together, our results reveal that it is possible to strengthen the resilience of education systems, enabling education provision amid disruptions, and to deliver cost-effective learning gains across contexts and with governments.

More than 2 billion people live in countries that are affected by emergencies that frequently disrupt education. Although they are frequent and disruptive, education emergencies have historically been understudied. Here we present a new database that documents just how frequent and disruptive such shocks can be. Figure 1 documents the extent of school disruption for a selected set of emergencies over the past two decades.

Schools close for long periods during these emergencies, and learning loss can be substantial[2–5]. Education Cannot Wait, the United Nations' Global Fund for education in emergencies, estimates that 222 million children experience regular schooling disruptions and are in active need of education in emergencies programmes[1]. In low-income countries, students lose on average 10% of the academic year owing to climate shocks alone[6]. Moreover, school breaks, although they are not emergencies, are another regular school disruption, and often results in substantial learning loss among low-resource households[7,8]. These shocks exacerbate a pre-existing learning crisis, leaving many students well behind grade-level expectations[9]. Fewer than half of primary-age students in low- and middle-income countries are able to read a story or perform two-digit mathematical operations[10].

Resilient education systems need to promote learning across settings and to withstand frequent disruptions. Ideally, interventions that promote learning in emergencies should be low-cost, simple to implement and quick to deploy. They should also yield high take-up across different geographies and implementation models and should be targeted to children with diverse educational and cultural backgrounds. Understanding which approaches can be effective across these circumstances requires multi-context, multi-model studies. However, rigorous evaluation of approaches to deliver education in emergencies remains challenging and rare, especially across contexts and with governments.

Here we evaluate a set of education in emergencies programmes to promote learning during large-scale school disruptions, which affected more than 1 billion children worldwide. We conducted five randomized controlled trials, including multiple delivery models, such as NGO teacher aides and government teachers, to test scalability by government systems. Our trials were conducted across five countries: India, Kenya, Nepal, Uganda and the Philippines. In all settings, schooling was disrupted owing to the COVID-19 pandemic, and several of the countries in our study experienced some of the longest school closures in the world.

We use mobile phones to provide various educational interventions to primary school children. Mobile phones provide a platform that can reach households cheaply at scale in low-resource contexts[11]. Whereas less than 15% of households have access to the internet in low-income countries, more than 70% have access to mobile phones[12]. Moreover, mobile phones enable teachers to reach students at home even when school is disrupted, providing a resilient and flexible modality to provide education during emergencies. Mobile phones are increasingly relied on to provide aid via mobile money during humanitarian crises[13], yet

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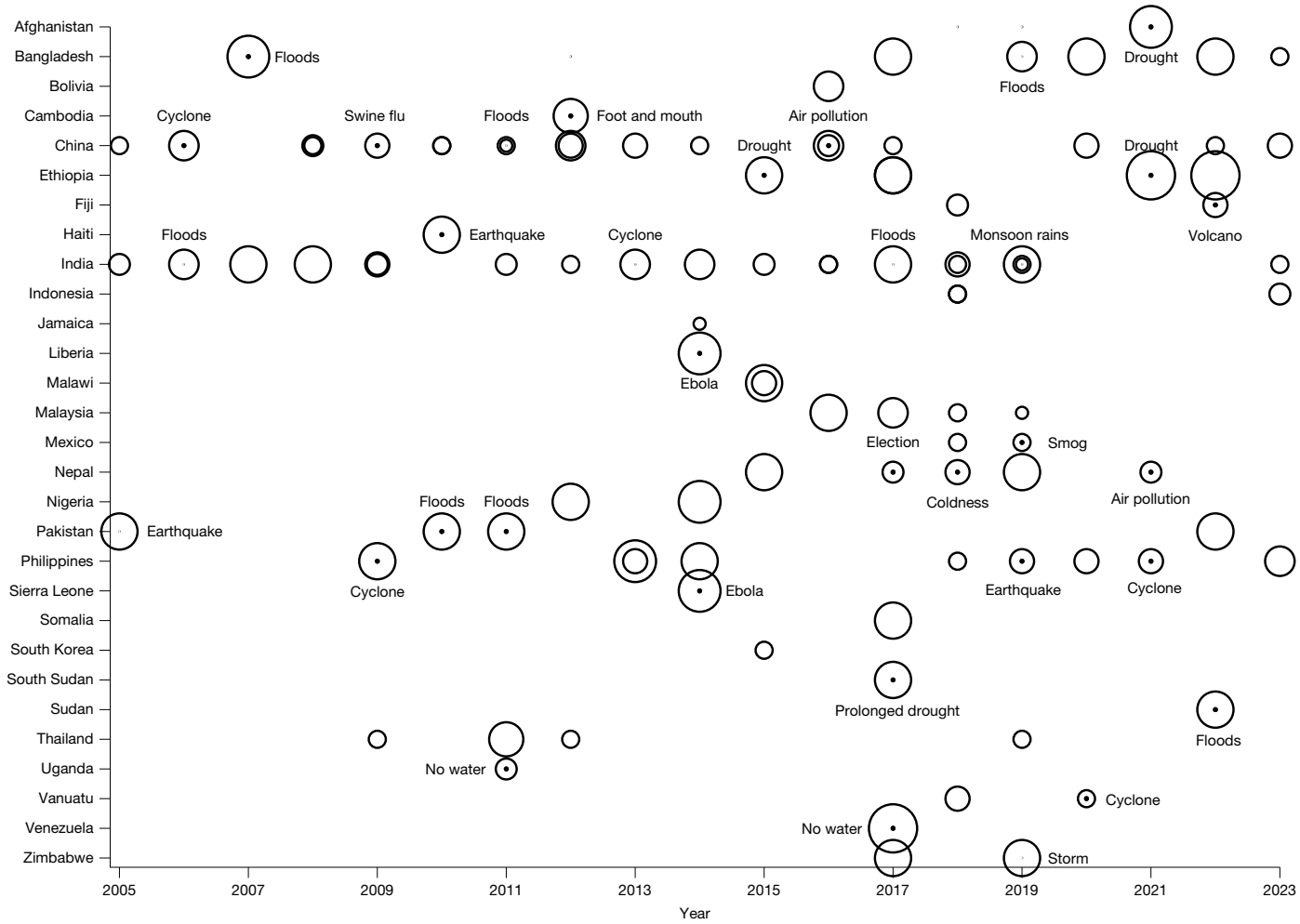


Fig. 1 | Documenting a set of education in emergencies between 2005 and 2023. The graph plots an index of the length of school closures and number of people affected by shocks that have disrupted schooling by country and year. The size of each bubble increases with the duration of school closure, the number of people affected, or both. Thicker or multiple circles represent multiple events in the same year and country. More information on the data compilation is available in the Supplementary Information. The data comprise compiled school closure information based on press releases of the United Nations Office for the Coordination of Humanitarian Affairs (OCHA) ReliefWeb, World Vision, United Nations Children's Fund (UNICEF), the British Broadcasting Corporation (BBC) and other local outlets. Major disruptions were also identified through The International Disaster Database (EM-DAT).

remain underutilized to provide mobile education. One treatment included a set of SMS messages, such as numeracy content provided weekly, as well as nudges to engage in educational activities. A second treatment provided additional weekly 20 min phone call tutorials for 8 weeks. The educational pedagogy was as essential as the mobile phone platform. Phone calls covered foundational numeracy and aimed to target instruction to student learning levels via regular low-cost, high-frequency assessments. This builds on both in-person and technology-based targeted instruction approaches[14–19].

A proof of concept in Botswana showed that phone call tutorials were effective in promoting learning during initial pandemic school disruptions[20]. However, questions remain on whether this approach can be scaled across contexts and when delivered by governments—a pervasive challenge for social programmes[21,22]. To ensure scientifically robust findings, we need evidence across diverse settings and delivery models[23]. By conducting large randomized trials across five contexts as well as comparing models, such as NGO and government delivery, we contribute new evidence on effectiveness across diverse contexts and delivery models, key elements of scaling science[24].

This study contributes to the literature in several areas. First, we contribute to the nascent experimental literature on education in emergencies. Substantial research has been carried out in this area, but much of it has been qualitative or with small samples. One exception is a randomized trial in which an NGO provided schooling in rural areas of conflict-affected regions in Afghanistan and found large effects on learning and closing of gender gaps[25]. We expand this literature by evaluating alternative, scalable forms of education beyond traditional in-person schools, such as remote learning, which is the only option during many emergencies when schools close. There is a robust literature documenting the cost of school disruptions due to teacher strikes[26], earthquakes[2], school holidays[7] and COVID-19 (refs. 4,27–31). However, there is less evidence on effective approaches to stem these learning losses. We contribute evidence on scalable solutions to stem learning losses. A growing literature explores the effectiveness of remote tutoring, yet the results remain mixed[32–35]. Without a coordinated cross-country trial it is impossible to know whether differences in effects are due to the intervention or to the setting, since both vary simultaneously. For example, a recent study[33] tested a related but different intervention in Sierra Leone—using a similar mobile phone platform but deploying a different pedagogy—and found no effect of mobile phone tutoring. In their study, the tutoring call was linked to a uniform national radio programme rather than targeting instruction based on student learning week-on-week—a key difference. To assess whether the approach works across contexts we cannot simply aggregate prior

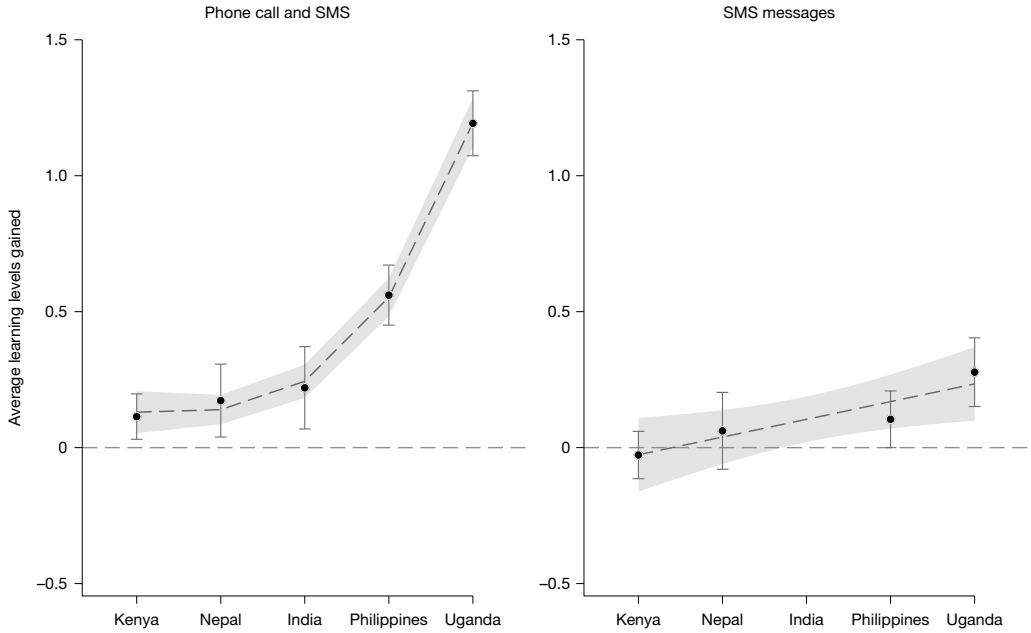


Fig. 2 | Main learning results across countries and treatments. Average raw unit operational learning levels gained in treatment groups relative to control groups. Treatment effects on learning outcomes are reported by country and treatment group with 90% confidence intervals within each country. Countries are ordered by the timing of the study, with Kenya first and Uganda last. The curve represents a fractional polynomial fit (phone arm) and a linear fit (SMS arm) of average learning levels gained against the order of the country in the study, each with a 90% confidence band. 'Learning levels gained' refers to the number of operations gained, with each operation scored as an additional level gain (beginner, 0; addition, 1; subtraction, 2; multiplication, 3; division, 4). Data are presented as treatment effects relative to the control group with ±90% confidence intervals. Sample sizes: India (n = 708), Kenya (n = 1,985), Nepal (n = 2,678), the Philippines (n = 2,469) and Uganda (n = 1,308). Further details on related results are presented in Extended Data Table 1.

studies in which both intervention and context vary—we need to hold the intervention constant, while varying sites in a coordinated multi-site trial. We conduct such a coordinated five-country trial in this study.

We also contribute to a growing literature on external validity and scale. Recent examples show the extent of the scaling challenge, with many social programmes that initially worked in proof-of-concept studies no longer delivering impacts when scaled across settings or when delivered by governments[22,24,36,37]. In contrast to some existing studies, our results identify an education approach that appears to work well across contexts and when delivered by governments. One reason might be that we leverage a particularly scalable technology: mobile phones[11,38]. A second reason may be the generalizability of the underlying mechanisms tested in our study, which relate to other best practices in education. For example, tutoring has been shown to be one of the most effective, although expensive, approaches to improving learning in high-income settings[39,40]. The phone call tutorials provide a cheaper version of tutoring that is applicable in low- and middle-income contexts. In addition, since the phone calls are one-on-one and have frequent learning assessments, they enable targeted instruction to every child's level, another approach shown to consistently improve learning[15–17,19]. Our study further advances the external validity and scale literature by conducting randomized trials across five countries in a literature where very few randomized controlled trials are multi-country studies. We also advance the scale literature by introducing experimental evaluation both of government relative to NGO delivery as well as randomizing those teachers who implement to evaluate effects on ongoing teacher practices. These experimental variations enable us to assess questions of effectiveness within government systems.

Finally, we contribute to the global education literature, with a focus on improving learning outcomes. Over the past several decades, education enrolments have improved worldwide, yet learning has barely budged[10,41,42]. Growing evidence reveals that popular input-only reforms, such as general teacher training, provision of computers, or school grants, are not enough to improve learning. By contrast, approaches that improve the quality of teaching, such as teaching at the right level and structured pedagogy, can generate large improvements in learning[43–46]. The learning gains from the phone-based tutorials tested in this study can deliver up to 4 years of high-quality schooling per US$100 (ref. 47). These types of highly cost-effective approaches can more easily scale and help address a persistent global learning crisis in low- and middle-income contexts.

Main results

We report results on a set of rich, harmonized data across five countries using both baseline and endline surveys. Questions included a learning assessment with the child and several additional child indicators, such as well-being, self-perceptions and non-cognitive skills. We also collected parental beliefs and behaviours. A portion of the survey was conducted with the parent, and learning outcomes were collected by directly assessing the child over the phone. A set of common questions were included in the baseline and endline surveys and across countries for comparability.

We report results from large-scale randomized controlled trials across five countries: India, Kenya, Nepal, the Philippines and Uganda. In Nepal and the Philippines, we include government and NGO delivery treatment arms. All countries except India also included a comparison SMS-only treatment relative to a phone call plus SMS treatment. Randomization enables identification of causal effects of the education in emergency interventions studied. Further details on the data and experimental methods are provided in the Methods section. Further details on the contexts and interventions are provided in the Supplementary Information.

Our main results are shown in Fig. 2 and show positive effects for the combined phone and SMS treatment in all countries, with increasing effectiveness over time as the trials progressed; for the SMS-only treatment, effects are more mixed across countries. Effects are further elaborated on in Extended Data Table 1, where we report effects

across contexts on the number of operations gained and the highest proficiency attained. We also show pooled average effects across all five countries. Column 1 in Extended Data Table 2 shows that for the combined phone and SMS group, there was a 0.321 s.d. ($P < 0.001$) increase in the average numerical operation across all 5 countries. SMS effects are smaller on average, with a 0.078 s.d. effect ($P = 0.006$). As shown in Fig. 2 and Extended Data Table 1, SMS messages only have a detectable statistically significant effect in a subset of contexts (Uganda and the Philippines). These results suggest that SMS messages can work in contexts with the largest need. However, live phone calls may be necessary to strike the balance of being intensive enough to deliver impact that can be sustained and scaled across diverse contexts while remaining cheap and scalable.

In addition, although effects of phone call tutorials are consistently positive, they vary in magnitude. Results are largest in later trials and in countries that experienced the longest school closures: Uganda and the Philippines. As shown in Extended Data Table 1, students in Uganda gain more than a full operation level (1.193 levels, $P < 0.001$). This translates to substantial proficiency gains, with 34.4 percentage point gains ($P < 0.001$) on the highest proficiency level. In the Philippines phone tutorials yield 0.56 level gains ($P < 0.001$) and a 13 percentage point increase ($P < 0.001$) in students reaching the highest operation level. In both countries, students were out of school for almost 2 years and the counterfactual for the control group was very limited schooling. These countries were also the later trials, with implementation fidelity improving over time. We discuss both of these mechanisms in more depth later on. Results are robust to the inclusion of a suite of control variables.

Mobile phone calls saw high take-up and high engagement throughout the programme, consistent with high impact. Detailed monitoring data shows high week-on-week engagement across sites ranging from 70% to 80%, as shown in Extended Data Fig. 1, even amid disruptions and across diverse low-resource settings. This highlights that the programme reached households using a platform that they found easy and convenient to access on a regular basis. This contrasts with the low rates of take-up of other platforms, with few households typically taking up alternative delivery modes such as online, television or radio educational resources.

For both the SMS group and the combined phone and SMS group, the magnitude of the impact and statistical significance are robust to different estimation approaches. In Extended Data Table 2, we include baseline controls and country and grade fixed effects (column 2), and weight results by country arm (columns 3 and 4). We also adjust for multiple hypothesis testing in Supplementary Table A5, including sharpened q values to address the false discovery rate[48]. Since we have one outcome and two treatment arms, we consider two joint null hypotheses simultaneously. Estimates remain consistent.

Extended Data Table 3 and Supplementary Table A7 show learning outcomes across several proficiencies. Results show that the combined phone and SMS arm increases the share of students who get division problems correct by 13.5 percentage points ($P < 0.001$)—a 92% increase in division (from a control mean of 14.6%). These results by proficiency demonstrate that learning improved substantially in absolute terms, in addition to s.d. gains, and across a range of proficiencies. We also find increases in the share of students who are able to perform higher-order competencies, such as fractions (6.4 percentage point gains, $P < 0.001$). Since these competencies were not directly taught during the intervention, this reveals learning extended beyond familiarity with the content taught. These results further reveal potential dynamic complementarities in skill formation, showing that benefits of learning basic numeracy accrue to learning additional higher-order skills[49].

These results are large relative to the effectiveness of a typical education intervention. A recent review found that the median effective intervention yielded 0.10 s.d. gains in learning[50]. Moreover, a review of 150 interventions found that more than half of education interventions do not work at all[47]. These reviews put our results in context, with phone call tutorials being three times as effective as the median education intervention. Moreover, effects are even larger in these multi-country studies than in the Botswana proof-of-concept study, where learning improved by 0.12 s.d. (ref. 20). This result contrasts with prior literature showing that proof-of-concept studies rarely scale successfully to new contexts[21] or experience diminishing returns[37]. Rather than finding diminishing returns, we find that results improve as the approach is adapted, scaled and tested across contexts. We explore potential explanations later in the paper, including learning from experience as well as higher need for the intervention over time.

Table 1 | Learning outcomes for government delivery

	All government arms	Nepal	Philippines
	(1) Operations gained	(2) Operations gained	(3) Operations gained
SMS messages	0.048	0.072	0.060
	(0.049)	(0.080)	(0.061)
	[0.326]	[0.367]	[0.324]
Phone call and SMS: NGO	0.296***	0.132	0.510***
	(0.057)	(0.088)	(0.073)
	[0.000]	[0.134]	[0.000]
Phone call and SMS: government	0.399***	0.213**	0.629***
	(0.061)	(0.087)	(0.086)
	[0.000]	[0.015]	[0.000]
Observations	5,147	2,678	2,469
Control mean	1.527	1.689	1.388
P value: NGO versus government	0.110	0.356	0.209
Country fixed effects	Yes	No	No
Countries included	Nepal, Philippines	Nepal	Philippines

Learning outcomes comparing implementation delivery models (NGO and government). Learning refers to how a child scores on four basic numeracy options: no operations correct, addition, subtraction, multiplication and division (measured on a scale of 0–4). This variable takes the value of 0 for students who got zero operations correct, 1 if students got addition correct, 2 if subtraction was correct, 3 if multiplication was correct, and 4 if students got division correct. Column (1) pools both countries in which we tested implementation scale-up in government systems. Column (2) shows the results for Nepal alone, and column (3) shows results for the Philippines alone. Standard errors are in parentheses and *P* values are in square brackets. *$P < 0.10$; **$P < 0.05$; ***$P < 0.01$.

In Table 1, we show results for the subset of countries (Nepal and the Philippines) where we randomized delivery models to test scalability within government systems. Column (1) shows the pooled results: both government teachers and NGO instructors are effective at improving student learning, with government teachers improving learning by 0.40 operation levels across both contexts, and NGO teacher aides improving learning by 0.30 operation levels ($P < 0.001$ for both). These results demonstrate that education in emergencies programmes can be effectively implemented by government teachers, and not only by NGO implementers.

These results show substantial effectiveness when delivered by governments. Prior literature has found that programmes often work when delivered by NGOs but are not replicated when scaled by governments. For example, in Kenya, contract teachers improved learning, but when delivered by the government, the programme effectiveness waned[36]. A large-scale government teacher training programme in Nepal similarly found no effect, largely owing to poor implementation[51]. Our results build on this literature, providing an alternative view: government delivery can achieve large learning gains. This is consistent with other studies that have refined interventions over time to facilitate government adoption[15,17].

Heterogeneous treatment effects, shown in Table 2, reveal that the programme is broadly effective for students across the distribution of

Table 2 | Heterogeneity

	Caregiver education	Gender	Baseline level
	(1) Operations gained	(2) Operations gained	(3) Operations gained
SMS messages	−0.013	0.136***	0.277***
	(0.148)	(0.052)	(0.077)
	[0.933]	[0.008]	[0.000]
Phone call and SMS	0.288**	0.427***	0.219**
	(0.139)	(0.047)	(0.095)
	[0.038]	[0.000]	[0.021]
Primary	0.026		
	(0.110)		
	[0.812]		
Secondary plus	0.446***		
	(0.103)		
	[0.000]		
SMS messages×primary	0.079		
	(0.169)		
	[0.643]		
SMS messages×secondary plus	0.114		
	(0.154)		
	[0.462]		
Phone call and SMS×primary	0.275*		
	(0.160)		
	[0.085]		
Phone call and SMS×secondary plus	0.112		
	(0.145)		
	[0.439]		
Is female		0.093*	
		(0.048)	
		[0.052]	
SMS messages×is female		−0.070	
		(0.070)	
		[0.321]	
Phone call and SMS×is female		−0.024	
		(0.064)	
		[0.703]	
Baseline level			0.360***
			(0.050)
			[0.000]
SMS messages×baseline level			0.007
			(0.038)
			[0.848]
Phone call and SMS×baseline level			0.006
			(0.035)
			[0.857]
Observations	8,028	9,148	9,148
Control mean	1.726	1.726	1.726
P value: SMS versus phone call plus SMS	0.048	0.000	0.627
Countries included	All 5	All 5	All 5
Country fixed effects	Yes	Yes	Yes

Heterogeneous treatment effects on learning. Column (1) shows the results between having caregivers who reached primary education or less and caregivers who reached secondary education or more (secondary plus). Column (2) shows the results between boys and girls. Column (3) shows results across baseline learning levels including country-specific baseline level interactions. To enable estimation for the full sample of endline learning results, baseline learning is imputed for the random sample, and not assessed based on the average. Standard errors are in parentheses and *P* values are in square brackets. *$P < 0.10$; **$P < 0.05$; ***$P < 0.01$.

baseline learning levels. This relatively even impact across students' starting conditions, or characteristics such as baseline learning levels or gender, is consistent with the mechanism of teaching at the right level. Targeted teaching is designed so that instruction meets children where they are regardless of their grade or age or other characteristics. Thus, broad-based learning gains are likely to be due to the fact that the programme was highly and effectively targeted, benefiting children across the distribution. We also find slight heterogeneity by parental education. Column (1) in Table 2 suggests that the programme worked particularly well for students where the caregiver had lower levels of formal education (primary education or less). This suggests that results are strongest when there are fewer alternative education support systems at home. This also reveals that even in low-literacy contexts, parents can be effective conduits for quality instruction.

We embedded an additional randomized study allocating a random subset of teachers to deliver the phone call tutorials in Nepal out of a list of hundreds of eligible government teachers. This additional experimental variation enables detection of the impact of delivering the programme on teachers' beliefs and practices—a measure of potential spillover effects into the education system that can persist through teachers beyond the intervention. In addition, teacher randomization enables generalization to the broader set of eligible government teachers. Thus, results capture government delivery impacts which do not hinge on implementers being highly selected, and effects are likely to translate to typical government teachers in the education system.

Table 3 shows that teacher practices shift; teachers are 9.3 percentage points more likely to target their feedback to students' learning level ($P < 0.05$). Teachers are also more likely to get parents involved in education ($P < 0.1$). We further find large effects on teacher perceptions that they were able to help students learn, as well as their desire to teach, with a 15.8 percentage point gain in wanting to be a teacher if they could make the choice again ($P < 0.1$ and $P < 0.01$, respectively). Although these results are reported by teachers and should be interpreted with caution, they suggest that delivering effective programmes could unlock a virtuous cycle within government education systems, in turn motivating teachers to want to teach and to improve their teaching practice further.

We examine counterfactual learning trajectories in Extended Data Figs. 2 and 3 to contextualize gains in the treatment groups. In Uganda, where effects are largest, about 17% of grade 4 students can divide at baseline in the control group. At endline, only 10% can divide, showing substantial learning loss. In the treatment group, 48% of grade 4 students can divide at endline, fully recovering learning loss due to school closures. Moreover, only 21% of grade 5 students in the control group at baseline can divide, revealing that grade 4 students in the treatment group surpass grade 5 students in the status quo. Thus, not only does the intervention facilitate full recovery of learning loss, it exceeds typical learning trajectories by nearly an order of magnitude. In other countries, learning does not deteriorate, but simply increases slowly. The intervention accelerates learning in these contexts relative to the counterfactual learning trajectory.

In the Supplementary Information, we include several additional results. We analyse effectiveness during another education emergency—a typhoon that struck the Philippines during the study. We explore additional margins of impact, including parent beliefs, student beliefs and non-cognitive skills. Finally, we explore the potential for crowd-out, to test whether tutoring time displaces alternative productive activities. Overall, we find continued effectiveness during an additional education emergency, several margins of impact beyond learning outcomes, and limited crowd-out.

Mechanisms across contexts

We explore variation in effects of phone call tutorials, which worked consistently but heterogeneously, across diverse contexts. Analysing mechanisms to explain effectiveness across contexts is challenging, and

Table 3 | Changing teacher beliefs and practices

	Teacher practices		Teacher beliefs	
	(1)	(2)	(3)	(4)
	Get parents involved	Targets to student level	Would be teacher again	Helped students' learning
Teacher implemented call	0.088*	0.093**	0.158***	0.106*
	(0.050)	(0.046)	(0.045)	(0.058)
	[0.076]	[0.043]	[0.000]	[0.067]
Observations	290	290	290	290
Control mean	0.721	0.769	0.735	0.408

Treatment effects on teacher practices and beliefs. This includes government teachers in Nepal who were randomly assigned to implement the phone call programme. We successfully spoke to 83% of the government-employed mathematics teachers who were in the schools identified by local governments. Of the eligible grade 3–5 mathematics teachers that we spoke to, 81% expressed interest in participating in the study. Of this eligible pool of 301 government teachers, 50% were randomly selected to implement the programme. The endline response rate was 96% and was balanced across the treatment and control groups. Column (1) shows effects on teachers' beliefs that they can get parents involved in their child's education. Column (2) shows effects on teachers saying that they tailor student feedback based on students' individual understanding and skill level. Column (3) shows effects on teachers saying if they had to choose their profession again, they would still be a teacher. Column (4) shows effects on teachers saying they believe they helped their students with their mathematical skills. These are all dummy variables that are coded as 1 if respondents strongly agreed on a Likert scale. Standard errors are in parentheses and *P* values are in square brackets. *$P < 0.10$; ** $P < 0.05$; ***$P < 0.01$.

should be done with caution, since multiple factors vary simultaneously. Table 4 includes a summary of key facts per country trial. We conduct exploratory analysis using suggestive descriptive data and observable systematic patterns. We focus on two mechanisms with particular variation across contexts: rates of targeted instruction and counterfactual conditions. Various additional margins vary across contexts, but do not vary systematically or consistently enough to facilitate credible comparisons. For example, in Kenya, younger students were taught but this varies only once, rather than systematically across settings.

Platform and pedagogy

In addition to leveraging a scalable platform—mobile phones—the programme pedagogy of targeted educational instruction, often referred to as 'teaching at the right level', was a core dimension of implementation fidelity. Phone call tutorials were designed to target instruction to children's learning levels. We find that variation in effect across trials is consistent with variation in targeted instruction. Although results show that phone calls were consistently effective across countries, the magnitude of effects increases in tandem with the order of the trial: Kenya, followed by Nepal, India, the Philippines and finally Uganda. This order of effectiveness tracks the degree of targeted instruction. Detailed monitoring data provides evidence of increasing implementation fidelity as the trials progress. For example, the accuracy of targeted educational instruction increases from a starting point of 50.9% of students in Nepal to 81.5% on average in Uganda[52] (Extended Data Fig. 4). These data reinforce the importance of targeted instruction. As the targeting mechanism improved study after study, this coincided with the programme becoming more effective. These results suggests the 'voltage drop' in effectiveness over time[21]—all too common when scaling—is not inevitable. Equipped with monitoring data and coordinating mechanisms to learn across trials, these results show that programmes have the potential to experience a 'learning curve' with improved fidelity (such as having ever more targeted instruction) and thus higher impact over time as they are scaled and implemented in new contexts. In addition to the trial-by-trial improvement in targeting accuracy, we find high rates of targeting accuracy through both NGO and government delivery. This consistent targeting accuracy across implementer types may explain the high effectiveness of both NGO and government teachers. It also adds evidence regarding the importance of targeting accuracy as a mechanism for impact[52].

These data suggest that effective targeting pedagogy has an important role in programme effectiveness. This mechanism may explain why the approach tested worked across diverse contexts. Consistent with the importance of these mechanisms, a phone education programme that did not rely on these principles had less impact: phone calls in Sierra Leone that delivered non-targeted lessons based on a mass radio programme to students did not improve learning[33]. In the Sierra Leone study, the approach used an accessible platform without targeted pedagogy. To successfully scale across different contexts, ensuring that targeted instruction is in place is likely to be key to improving learning.

Counterfactual conditions

Extended Data Table 4 shows several statistics characterizing counterfactual conditions. We observe clear patterns that show more severe counterfactual conditions coinciding with larger impacts. In Uganda and the Philippines, the settings with the highest impacts, we also see the most severe conditions, with more disruption and fewer alternative educational instruction options. Uganda and the Philippines have the highest degree of disruption in general and during the study, experiencing up to 2 years of disruption, with disruption occurring during the entire study period. They also have the largest share of lagging students, with 9% or less of students reaching the top operation level in the status quo. Similarly, we see that Uganda and the Philippines have low rates of outside options, with less than 22% of households saying they received any calls from educators in the past few weeks.

Although these counterfactual conditions do not follow an exact rank order with effect size, the broad patterns are consistent—more severe conditions map broadly to larger effect sizes in Uganda and the Philippines. This suggests the interventions studied are likely to add value particularly in the most disrupted education in emergency settings.

Cost-effectiveness

An important feature of the education in emergencies approaches tested in this study is that they have low cost. The primary tool required to implement the programme is a mobile phone, a device that most households have access to, even in low-resource settings[53]. Since the approach builds on existing household infrastructure, the main costs are related to content delivery and connecting with families, which are often marginal, such as airtime for phone calls. In addition to being low cost, the approach has low procurement needs, an attractive feature for governments.

We carefully collected cost data in each trial. Our estimates suggest an average cost per child of the phone call and SMS tutorials of about US$11 per child. This estimate is even cheaper than estimates in the proof-of-concept study in Botswana owing to economies of scale as the approach has scaled up, including cost savings such as use of existing pedagogical material, shorter and more efficient training, and streamlined data monitoring systems. We benchmark the impacts of the programme against other education programmes using a variety of approaches. First, we compare raw estimates of effectiveness and cost with similar programmes such as tutoring and targeted instruction. Phone call tutorials simulate the benefits of one-on-one tutoring, which has been shown to be one of the most effective educational approaches[39]. However, many tutoring programmes have high cost. For example, a prominent tutoring programme yielded 0.19 to 0.31 s.d. learning gains at a cost of US$2,500 per child. In low- and middle-income settings, phone calls could provide similar or larger impacts two orders of magnitude more cheaply, enabling scale-up across diverse settings. The SMS-only treatment, when effective, can be extremely cost-effective, at 41.1 learning-adjusted years of schooling (LAYS) per US$100. However, whereas phone and SMS effects were consistent across all countries, the effects in the SMS-only arm varied across contexts. Given this, it seems

Table 4 | Trial description

	India	Kenya	Nepal	Philippines	Uganda
Sample size	850	6,724	3,732	3,492	2,138
Randomly selected endline sample size	765	3,556	3,351	3,164	1,495
Student grades	3–5	1–2	3–5	3–4	3–5
NGO delivery	✓	✓	✓	✓	✓
Government delivery	—	—	✓	✓	—
Unit of randomization	Household	School-grade cluster	Household	NGO: household; government: school-grade cluster	Household
Stratification variables used in block randomization	Student baseline level, school, female	School	Local government, parent perception of student baseline level	NGO: region, student baseline level; government: school, grade, student baseline level	Student baseline level, attends government school, previous education programme participation
Included combined phone call and SMS treatment	✓	✓	✓	✓	✓
Included SMS-only treatment	✗	✓	✓	✓	✓
Implementer type	NGO (Alokit)	NGO public–private partnership (NewGlobe)	Government (Ministry of Education), World Bank, NGOs (Teach for Nepal, Street Child)	Government (Department of Education), research NGO (Innovations for Poverty)	NGO (Building Tomorrow)
Same teachers as school instruction	—	✓	—	✓ – government delivery	—
Number of weeks of implementation	8	8–12	8	8	8
Dates	Apr 2021–Jul 2021	Dec 2020–Apr 2021	Jan 2021–Jul 2021	Aug 2021–Jul 2022	Oct 2021–Jan 2022
Administrative units in study	1 state	30 counties	All 7 provinces	3 regions	9 districts

Summary of key features of the intervention across countries. Implementers of the interventions included government education ministries and NGOs.

plausible that the SMS-only intervention represents a cost-effective option in extreme education emergencies where, for example, calls are not an option or schooling is disrupted for such an extended period of time such that any provision of content is substantially better than the status quo.

Second, we use LAYS, a cost-effectiveness measure in education that has been estimated for over 150 impact evaluations in low- and middle-income countries; each LAYS can be interpreted as a high-quality year of schooling gained. We find the programme yields 3.6 LAYS per US$100, ranking in the top 10 out of 150 education interventions reviewed[47]. This result highlights the potential for phone call tutorials to deliver value to education systems and students in a broad array of contexts, both during and potentially even outside education emergencies. Even before the COVID-19 pandemic, there was high demand for cost-effective education programmes to help address the global learning crisis. Our results suggest phone-based targeted tutoring programmes, such as the one tested in this study, have the potential to deliver cost-effective learning gains across contexts and with governments.

We also contribute new evidence on the robustness of a cost-effective approach to measure learning—high-frequency phone call assessments. We leverage data across five countries and a series of randomized validity tests. Extended Data Table 5 shows the results of five checks we conducted, described in more depth in the Methods section. Results suggest that measurement of learning by phone can be highly robust, in addition to being cost-effective.

Conclusion

Here we present results from large-scale randomized trials evaluating the provision of education in emergency programmes across five countries: India, Kenya, Nepal, the Philippines and Uganda. We test multiple scalable models, including government delivery, of remote instruction for primary school children during COVID-19, which disrupted education for more than 1 billion schoolchildren worldwide, as well as during an additional emergency—a typhoon—which further disrupted schooling.

Despite heterogeneous contexts, the results show that the effectiveness of phone call tutorials can scale across contexts; we find consistently large and robust effect sizes on learning, with average effects of 0.30–0.35 s.d. In the subset of trials in which we randomized whether the intervention was provided by NGO instructors or government teachers, we find large effects for both delivery models, indicating effectiveness when delivered within government systems (Supplementary Fig. A1).

These results have relevance to global efforts to support education in emergencies including conflict, disease, natural disasters and climate shocks, which routinely shut schools down. During these shocks, alternative models are needed to deliver education. The results presented here identify approaches that can scale effectively during disruptions across contexts and cost-effectively improve learning for students. This study further demonstrates that rigorous testing of programmes in humanitarian settings is possible.

Even outside education emergencies, millions of children worldwide learn very little in school, either because they are taught curricula beyond their learning level, or because they are unable to access quality instruction since they live in remote areas. Given widespread mobile phone ownership rates globally, phone-based tutoring programmes such as the one studied here have the potential to maintain schooling continuity and accelerate learning even outside emergency settings or after the initial emergency has eased. The low cost, high access and ease of implementation of mobile phone tutoring could build more resilience into education systems, enabling systems to better withstand frequent shocks, and to more generally utilize cost-effective approaches to address a persistent global learning crisis. One high profile example includes Karnataka, India, which is scaling this approach—now called ConnectEd globally and referred to in Karnataka as Ganitha

Ganaka—during school and to prepare for school disruptions due to monsoon seasons. Another example includes follow-on trials that built on this study in Latin America[54].

This study has substantial policy implications. Multiple efforts to scale the intervention studied in this paper are underway worldwide. This paper provides critical evidence to inform effectiveness across settings and delivery models. Future research could examine effects at even larger scale. As Extended Data Fig. 5 shows, the trials covered substantial portions of the geographies studied. However, there is still value in exploring effects at national scale in future work. Future research could examine refinements to the delivery model to engage parents further[55] and could evaluate potential spillovers on teacher instruction and student learning in the classroom. Moreover, additional work could explore long-term benefits following observed gains during the study period, with learning potentially compounding over time as skills beget more skills.

Online content

Any methods, additional references, Nature Portfolio reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at https://doi.org/10.1038/s41586-026-10990-x.

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Methods

Data

In each country, we have two waves of data: baseline and endline. In Nepal, we conducted a baseline survey with a random 50% of the sample, and in Kenya, we relied on administrative data instead of baseline survey data. The endline surveys took approximately 30 min to administer and included approximately 20 questions. These questions included a learning assessment, child wellbeing, parental engagement in educational activities, and parental perceptions of their child's learning. A portion of the survey was conducted with the parent, and learning outcomes were collected by directly assessing the child over the phone. Endline surveys were conducted a few months after the programme ended. A set of common core questions in the baseline and endline surveys are included in the Supplementary Information.

The learning assessment was adapted from the ASER (Annual Status of Education Report) test, which has been used frequently in the literature to measure learning outcomes[14,15] and is used routinely across 14 countries. The test consists of multiple numeracy items, including two-digit addition, subtraction, multiplication and division problems. In addition, we asked students to solve a word problem and a fraction problem to capture learning outcomes beyond a core set of mathematical operations. Results are presented in several main results figures and tables. Distributions of raw baseline and endline learning levels are shown in Supplementary Fig. A3.

To maximize the reliability of the phone-based assessment, we introduced a series of quality-assurance measures. To minimize family members in the household assisting the child, students had a time cap of 2 minutes per question and we asked each child to explain their work. We only marked a problem correct if the child correctly explained how they solved the problem and enumerators were confident parents were not assisting their child. We also conducted a battery of validity checks to ensure the reliability of the learning outcomes. These results contribute new evidence on the robustness of remote learning assessment data across five countries. Phone assessment has emerged as a common strategy for large-scale household surveys such as the World Bank Living Standards Measurement Survey (LSMS). A growing literature has started to explore the validity of phone-based assessments to measure learning outcomes[20,56–58], with emerging evidence suggesting phone assessment can capture meaningful information at high frequency and low cost.

Extended Data Table 5 shows the results of five checks we conducted on the validity of our main learning outcomes of learning assessments via phone. Column 1 shows the first robustness check where we compare in-person to phone-based assessment for a representative sample of the same students in Kenya. An additional test included back-checks, with a random subset of students tested twice on the same competencies. We find a strong relationship as expected, with large positive coefficients and *t*-statistics ranging from 5 to more than 20. We further randomize students to receive different problems of the same proficiency (for example, four different questions to measure two-digit addition with carryover). Finally, column 5 shows results from a real-effort question to disentangle effects of the intervention on effort on the test versus cognitive skills. Students were asked to answer several effort tasks, for example figuring out the day of the week or counting zeros and ones. Altogether, results suggest measuring learning by phone can be robust, similar to findings from the World Bank LSMS for household measures of consumption.

The survey also included questions on caregiver engagement in their child's education and beliefs. We measured engagement by asking caregivers how often they spent helping their child with their schoolwork over the previous weeks. We also included a measure of a caregiver's confidence in their child having made progress in learning over the previous months, and their perception of their child's numeracy level. Additional questions included information on whether the caregiver has returned to work. We also asked about parents' demand for remote learning services in the future, and whether they would be willing to pay for such a programme. For students, we asked about child's mental wellbeing, how much they enjoy school, and the child's own belief about what mathematics problems they will be able to answer. We also measured non-cognitive skills, such as perseverance and ambition. Results are shown in Supplementary Tables A2 and A3. Finally, we included demographic questions, recording the child's age, grade and gender.

The overall sample size, pooling all sites, is 16,936 households. The flow of participants from enrolment through randomization to endline follow-up is summarized in a CONSORT diagram (Supplementary Fig. A5). For endline surveys, we randomly sampled households from the full sample to interview for a total sample of 12,331. This was due to time and cost constraints. In Supplementary Table A8, we show that those randomly selected for endline interview are statistically equivalent to the full sample at baseline along a series of indicators such as gender, student grade and baseline learning level.

Supplementary Table A9 presents the response rate to the endline surveys and an analysis of survey attrition for those randomly selected to be part of the endline sample. The follow-up rate was around 76% of respondents at endline. Supplementary Table A9 also presents a test of whether response rates differed by treatment assignment. This provides evidence that our sample has a high and unbiased response rate.

Finally, we also include a survey for teachers to assess their beliefs and instructional practices. These questions include their desire to be a teacher, and teachers' view that the phone call tutorial programme was helpful for student learning. Questions also include instructional practices, such as involving parents in education further and better targeting feedback to students' actual learning level. These questions can capture potentially persistent effects on educational systems through teachers changing their beliefs and behaviours beyond the lifecycle of the programme.

Experimental design

We ran randomized controlled trials across five countries. Every trial had a control group and a combined phone call and SMS treatment arm. All studies except the trial in India also included an SMS-only treatment arm. In two countries, Nepal and the Philippines, we further randomized delivery of the phone call by NGO instructors or government teachers to assess scalability within government systems. We exploit random assignment to identify causal effects and estimate the impact of these education in emergencies interventions and various scalable delivery models.

We first estimate intent-to-treat effects per country *c* as follows:

$$Y_{i,j,c} = \alpha + \beta_1 \text{PhoneCalls}_{j,c} + \beta_2 \text{SMS}_{j,c} + \gamma X_{j,c} + \delta_s + \varepsilon_{i,j,c} \quad (1)$$

where $Y_{i,j,c}$ is a learning outcome for individual i in household j in country c. $\text{PhoneCalls}_{j,c}$ and $\text{SMS}_{j,c}$ take on the value 1 in their respective treatment arms and 0 otherwise. $X_{j,c}$ denotes a vector of baseline control variables, applied consistently across trials to enhance statistical power and precision: student grade, student gender and baseline learning level. For baseline learning, we have the same assessment as the endline in all countries except Kenya, where we use administrative exam scores, and in Nepal, where we use parent perceptions of baseline learning. Since random subsamples are drawn at baseline in a few cases, we complete the sample with average estimates to enable analysis to consider the full sample at endline when including controls. δ_s denotes the geographic strata fixed effects—the randomization stratum relevant to each country (for example, region, district or constituency). We assess impact on operations gained as well as the highest proficiency achieved in a given country. We estimate the regression using ordinary least squares. Our main specifications are

conducted per country. Given the speed and timing of the studies conducted during a historic crisis, rather than register new analysis plans, we rely on specifications published in prior related studies to guide our analysis[20]. Our specifications are straightforward, with one main learning outcome measure. In the extended data, we conduct corollary analysis where we pool across studies and we include country fixed effects η_s. We run additional regressions using alternative weights by country as well as country-by-country estimations. Our primary analysis focuses on learning outcomes expressed as operation levels gained in raw units as well as the proportion of students who make it to the top proficiency level. We analyse standardized learning outcomes only as a corollary analysis (standardized relative to control group standard deviations at endline and centred at mean zero). All tests are two-sided statistical tests. All analysis was conducted in Stata/MP 18 software.

We also ran a subset of trials in which phone calls were delivered by NGOs or government teachers. We estimate government (Gov) versus NGO delivery effects per country using the following specification:

$$Y_{i,j,c} = \alpha + \beta_1 \text{PhoneCalls}_{j,c,\text{NGO}} + \beta_2 \text{PhoneCalls}_{j,c,\text{Gov}} + \gamma X_{j,c} + \delta_s + \varepsilon_{i,j,c} \quad (2)$$

All standard errors are clustered at the level of the unit of randomization. In most countries this was the household level, which was de facto the same as the student level since one student participated per household; in a subset of treatment comparisons it was the school grade level. In India, Nepal and Uganda all randomization was at the household level. In the Philippines, the NGO arm was randomized at the household level; the government arm was randomized at the school grade level, with a corresponding control group randomized at the same level for each arm. In Kenya, the phone call arm was randomized at the school grade level whereas the SMS arms were cross randomized at the household level. Of note, robustness checks find little to no difference if standard errors are clustered or not across all specifications. This is largely due to the fact that there is substantial variation in learning outcomes within a cluster, such that the intracluster correlation is very low at about 0.03 in the Philippines and Kenya. The above specifications are the core estimation strategies. Additional specifications include other outcomes of interest beyond learning, such as engagement, as well as heterogeneity analysis, and various types of pooling across countries.

Due to randomization, we expect treatment and control groups to be balanced at baseline in expectation. Supplementary Table A10 presents balance tests using baseline data for key demographics and the same learning level variable. We find no statistically significant differences across multiple dimensions, including gender, grade, caregiver type, caregiver education levels and baseline learning. This is robust to multiple specifications, including with and without country fixed effects. Similar results are found for each country, with balanced samples in each individual country. This reveals that each randomization yielded balanced arms in line with expectations.

In all sites, the experimental design had high compliance rates; nearly all randomized treatment households participated in the expected treatment group. Detailed weekly monitoring data shows high take-up and fidelity, as shown in Extended Data Fig. 1.

In a secondary analysis, we embed another randomized trial which randomly allocated a subset of government teachers to deliver the phone call tutorials out of a pool of hundreds of eligible teachers in Nepal. Supplementary Table A6 shows balance tests with no statistically significant differences among treatment groups. We assess whether participating in phone call tutorials changed teachers' subsequent beliefs and behaviours, which we estimate for teachers *i* at the school level *j* in each country *c* as follows:

$$S_{i,j,c} = \alpha + \beta_1 \text{TeacherCalls}_{i,c} + \gamma X_{i,c} + \delta_s + \varepsilon_{i,j,c} \quad (3)$$

where $S_{i,j,c}$ captures teacher beliefs or instructional practices. These outcomes measure potential spillovers into the education system through teacher beliefs and behaviours which could persist beyond the programme.

Ethics statement

Insitutional review board (IRB) approval was granted by the Health Media Lab IRB (889OXFD21) and Columbia University IRB (21-384). Informed consent was performed in accordance with all relevant guidelines and regulations and obtained from caregivers, child assent and government approvals in each country. An additional country-specific IRB approval was granted by Nepal's Health Research Council (776/2020) due to local regulations. The trials were registered in the AEA registry: AEARCTR-0009779. After the study period, control group students were later able to be enroled in the intervention to benefit from the programme.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

Anonymized replication data are available at https://github.com/nangrist/mobile-education.

Code availability

All replication code is available from https://github.com/nangrist/mobile-education.

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Author contributions N.A. and C. Cullen conceived and designed the study. M.M., T.L. and C. Crossley provided technical intervention and partnership assistance and coordinated implementation across sites. Country leads led implementation and data collection in India (S.P.B.), Kenya (T.S.), Nepal (S.S.), the Philippines (R.M.P.) and Uganda (M.A.). N.A. and C. Cullen led the analysis. S.S. and P.B. contributed substantially to study design and interpretation. N.A. led the manuscript drafting and editing with input from all authors. All authors reviewed and approved the final manuscript.

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Competing interests N.A. and M.M. co-founded Youth Impact, the non-governmental organization that first developed and tested the ConnectEd phone tutoring programme evaluated in this study. C. Cullen heads Youth Impact's research team. Youth Impact provided training and technical assistance to implementing partners across the five trials but did not itself deliver the intervention in the study countries and received no financial benefit contingent on the study's findings. All other authors declare no competing interests.

Right-sizing Incentives

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Abstract

This paper presents results from a field experiment in Nigeria in which expectant mothers were offered cash payments of N2500, N4000 or N10000 conditioned on using formal medical care. A secondary intervention provided some mothers with an information signal about their pregnancy risk. We find highly non-linear effects of conditional payments—most of the incentive effect occurs on the extensive margin, with rapidly diminishing returns to incentive size. We also uncover new spillover effects: showing that incentives increase healthcare use in childhood by index children and by other children in the household. We find largely null effects of the information signal.

1. Introduction

Reducing child deaths in developing countries is a global development priority (United Nations, 2015; World Health Organization, 2018). As effective childhood interventions such as vaccination and diarrhea prevention have become ever more widely adopted, very early deaths—before and in the immediate aftermath of delivery—have accounted for an increasingly larger share of child deaths, underscoring the importance of early-life intervention.1 In a previous experiment, we showed that offering expectant mothers in a high-mortality setting small conditional payments to visit health facilities for care (henceforth in utero incentives) caused a large shift from the informal to the formal health sector, leading to a 25 percent decrease in child mortality (Okeke and Abubakar, 2020).2 Those findings suggested potentially large welfare gains from incentivizing mothers to use formal health care, despite well-known limitations within the formal sector (Das and Hammer, 2014).

This paper builds directly on those findings. We present new results from a follow-on experiment in which we randomized expectant mothers in Nigeria to be offered conditional cash payments of varying amounts—N2500, N4000 or N10000 (N360 = $1). We leverage this variation to empirically estimate the incentive response function. We examine how utilization changes in response, and more importantly we examine how child mortality changes in response, providing key missing evidence to policymakers. In the experiment we also tested a secondary intervention targeting informational constraints. Specifically, we provided new information to a random half of study participants about their pregnancy risk (based on observable risk characteristics). We examine how households respond to this information signal.

A total of 36,607 women participated in the trial. We interviewed all study participants at baseline and again following the conclusion of the pregnancy (about four and a half months after). We conducted this experiment between 2019 and 2020. We note this because the Covid-19 pandemic began about half-way through. Given its well-documented disruptive impacts, throughout this paper we will present separate results for the sample of mothers whose pregnancies were completed before the onset of the pandemic vs. after (pre/post-Covid samples respectively). We will show that behavior (and the response to the treatments) changed during the pandemic in sometimes unpredictable ways. We thus lean on the results from the pre-Covid sample. They are more indicative of what we might expect to find if we ran this experiment again.

Our first key finding is that the incentive response function is highly non-linear. We show that most of the effect of in utero incentives is concentrated on the extensive margin—between no and some incentive. Increasing the amount offered has a relatively small effect. To illustrate: going from N0 (no incentive) to N2500 nearly doubles the rate of health facility deliveries from 28 to 55 percentage points. Going from N2500 to N4000 (a 60 percent increase) buys an additional 1–2 percentage points. Going from N4000 to N10000 (a 250 percent increase) buys another three percentage points. In other words, a four-fold increase in incentive size buys only an additional 12 percent increase in utilization. We carry out multiple checks to show that this conclusion is robust. The mortality response is similar: fetal deaths (stillbirths) decrease by 1.2 percentage points in the N2500 arm (off a baseline of 5 percent), and by a nearly identical magnitude in the N4000 and N10000 arms, aligning closely with the utilization result. We find that these mortality effects largely disappear during the pandemic (i.e., in the post-Covid sample). Changes in utilization do not appear to explain these effects. We speculate that disruption in services and deterioration in quality may have reduced the returns to medical care during the pandemic (see for example Okeke, 2022).

Our second key finding is that in utero incentives generate substantial spillovers. We find that incentivizing mothers to use health care services during pregnancy ‘spills over’ to increase healthcare use after birth. We show that children born to treated mothers are more likely to be taken for a postnatal visit, and are more likely to be taken to a healthcare facility for treatment when sick (acute-care visit). We find similar non-linearities with respect to incentive size. Further, we show that this has meaningful health consequences. Diarrhea causes nearly 500,000 child deaths every year, nearly all of which can be prevented with a cheap and highly effective treatment—oral rehydration salts (Perin et al., 2022). We show that treated index children with diarrhea are about 11 percentage points more likely to receive oral rehydration salts. We also find spillovers to other children in treated households. We show that other young children in the household are also more likely to be taken for an acute-care visit.

Third, we find largely null effects of the information intervention. We find that providing an information signal to mothers about their pregnancy risk status does not lead to an increase in health care use, as we hypothesized. We find no effects on child mortality. Given our large sample size these null results are precisely estimated. We test whether information and incentives have complementary effects and largely find statistically insignificant results. There is a suggestion that information may have a small negative effect on utilization that is counteracted by incentives. Our analysis of data from a follow-up survey that we administered to mothers suggests that the information did not fundamentally alter women’s beliefs about their risk status, providing a potential explanation for its ineffectiveness.

Fourth, taking advantage of a second overlapping experiment that provided unconditional cash grants to some health clinics in the study areas—the grant loosened clinic financial constraints—we examine the potential interaction between conditional cash incentives, which target demand, and the supply side of healthcare delivery. Okeke (2026) shows that service delivery markedly improved in clinics that were selected for the grant. In this paper we test whether incentives have more pronounced effects when combined with supply-side improvements by comparing incentive/non-incentive sites with a grant/non-grant clinic. We theorize that incentive effects on health care utilization might be muted if access (or quality) is poor (Dipankar Rao and Peters, 2007; Aggarwal,

2021). Similarly, effects on health outcomes might be moderated by existing capacity (Godlonton and Okeke, 2016; Renard, 2022; Okeke, 2023). We find similar effects on utilization in sites with a grant vs. a non-grant clinic;3 however, we find larger mortality reductions in areas with a grant clinic, suggestive of complementary effects.

This paper adds to a growing literature on the welfare effects of in utero transfers (see Amarante et al., 2016; Okeke and Abubakar, 2020; Carneiro et al., 2021; Reader, 2023). Specifically, we find that in utero incentives reduce overall child mortality by about 15 percent—averaging across all incentive arms. To put this into context, Barham (2011) estimates that the Progresa program in Mexico reduced infant mortality by 17 percent among recipients; Rasella et al. (2013) find that the Bolsa Familia program reduced under-five deaths in Brazil (rate ratios of 0.94 for intermediate, relative to low, program coverage and 0.88 for high coverage); and more recently, Walker et al. (2025) find that a $1000 unconditional transfer to households in Kenya reduced infant mortality by 48 percent.4 Closer to this paper, Celhay et al. (2021) find that the Bono Juana Azurduy program in Bolivia, a conditional cash transfer program for pregnant women and children under 2 years old, increased survival rates for exposed birth cohorts by 3.8 to 14.7 percent. Finally, Okeke and Abubakar (2020), an evaluation of a similar incentive program, find mortality reductions of 18 to 26 percent.

This paper makes several important contributions: first, we provide novel evidence about how policy outcomes respond to variation in transfer size. While there is strong interest in the use of conditional payments to increase utilization of health care services specifically, and to improve household welfare more broadly (Richterman et al., 2023; Jacobs and Downey, 2022; Neelsen et al., 2021), little is known about how these outcomes respond to counterfactual changes in transfer size.5 This paper thus represents a major advance. Contrary to common wisdom we find that bigger is not necessarily better. The immediate significance of this finding is that policymakers, under the current status quo, may be leaving money on the table. Our findings suggest that we can extract additional welfare improvements simply by 'right-sizing'. The foregone gains are not trivial. To illustrate: Okeke and Abubakar (2020) find that a conditional payment of N5000 prevented one in four child deaths; the findings in this paper suggest that similar gains could have been achieved with a payment half as large. With a payment half as large, the program could have been offered to twice as many women, potentially saving twice as many children.

Our second contribution is to document the existence of previously unmeasured health spillovers. There is scant evidence of spillovers in the transfer literature. Guerrero et al. (2020) find that the Juntos program in Peru, which provided $30 monthly transfers conditional on targeted household members meeting schooling and health requirements, reduced the use of informal health care providers by targeted and non-targeted household members. Contreras Suarez and Maitra (2021) find that the Familias en Acción program in Colombia reduced illness reporting among non-targeted adults in treatment households. In line with these studies we find spillovers affecting non-targeted children. In addition we show that incentivizing particular health services also "spills over" to impact use of other (non-incentivized) health services.6 This is particularly salient given the well-documented link between healthcare use in childhood and later-life outcomes (Currie and Vogl, 2013; Bharadwaj et al., 2013).

Third, we present new evidence of the interaction between the demand and supply side. Others have speculated about this interaction (see for example Kruk et al., 2016; Das et al., 2018) but there has been very limited empirical evidence; in part because of the difficulty in finding plausibly independent variation in demand and supply. We have strong identification here because of the existence of two overlapping experiments: one targeting the demand-side, the other targeting the supply-side. We can distill our findings down to one simple message: providing incentives while simultaneously investing in improving service delivery leads to better outcomes. Each dollar spent on incentives generates a bigger return under better service delivery conditions. This is a message with intuitive resonance.

This paper also contributes to a large literature on informational interventions in health. Our findings are most relevant for the literature examining the impact of new information about one's health risk. Kim et al. (2019) conclude that providing risk information is, in general, not effective at inducing changes in behavior.7 They note: "…information alone is not necessarily sufficient to lead to behavioral changes and better health outcomes■" We concur. See also Edwards (2018) who finds zero treatment effects (on average) of notifying individuals about various health risk indicators (blood pressure, hemoglobin A1c, and cholesterol); Cook (2019) who find that being informed that one is overweight has no impact on subsequent weight, though receiving feedback that one is very overweight results in modest weight loss among higher-income individuals; and Zhao et al. (2013) who find that receiving information about being hypertensive (based on the results of a blood pressure test) leads to a reduction in fat intake, but only among the rich.

There is very limited evidence from developing country contexts. Godlonton et al. (2016) examine the effect of providing new information about the risk of HIV infection—circumcised and uncircumcised men are informed that male circumcision reduces the HIV transmission rate. They find some evidence that uncircumcised men who receive the information practice safer sex but find no effect on the demand for circumcision among adult men or male children. Baird et al. (2014) find largely no impact of HIV testing on health behaviors or school achievement in Malawi. We largely echo these findings. Importantly, we find that combining information with cash incentives largely does not alter this conclusion. This is a significant finding. Several other papers have found that adding information to cash transfers can increase impacts—see for example Fitzsimons et al. (2016) and Levere et al. (2024)—arguing that while information can correct knowledge deficits, cash provides the means to implement what has been learned. Our findings suggest that if the transfer is already conditional, then the information on risks does not increase impacts.

The rest of the paper is organized as follows: Section 2 provides some background and institutional details; Section 3 describes the data; Section 4 presents the results; Section 5 examines mechanisms; Section 6 examines the interaction between demand and supply; and Section 7 concludes.

2.4. Study interventions

There were two experimental household treatments: (i) a cash payment offered to the enrolled mother if she attended four prenatal visits and delivered the child in a health facility, and (ii) an information treatment that is described in more detail below. Randomization was at the site level, stratified by state, with half the sites assigned to each treatment. The treatments were crossed. We further randomized one-third of cash payment sites to one of three payment amounts: N2500, N4000, or N10000 (N360 = $1). The amounts were benchmarked against an existing government program (see Okoli et al., 2014).

Enrolled mothers were eligible for the cash payment, which was made after delivery, if they attended four prenatal visits and delivered the child in a health facility. Both conditions were required. To provide context about the magnitude of these incentives, we relate them to household spending. Based on data from the Nigeria General Household Survey, we estimate that the cash incentives of N2500, N4000, and N10000 correspond to about 7%, 11%, and 27%, respectively, of monthly household spending on food consumption (Nigerian National Bureau of Statistics, 2019). Using the cost of care in public health clinics as a reference point, we calculate that the payments of N2500, N4000, and N10000 represent a 76.3%, 122%, and 305% subsidy, respectively, relative to what households would pay on average for four prenatal visits and a facility delivery in the absence of the program.

Enrolled mothers were notified if they had any of the following risk factors based on age (15 years or younger or 35 and older), parity (first-time mother or five or more previous pregnancies), medical history (history of chronic disease, e.g., hypertension), and pregnancy history (a prior history of a stillbirth and vaginal bleeding during the current pregnancy).10 They were read a short statement notifying them and urging them to seek medical care during the pregnancy (see

Appendix B). If they did not have any of the assessed risk factors, they were read a short statement notifying them and also advising them of the importance of seeking medical care during pregnancy (see Appendix B). All other women were read a short statement about the importance of attending prenatal care and giving birth in a health facility.

Women with pregnancy risk factors ("at-risk" or "high-risk" women) are an important sub-group. The reason is simple: to the extent that the probability of a bad outcome can be mitigated, or at least managed, by medical technology, there is a strong economic rationale to target this group. Even if risk cannot be mitigated by medical technology, i.e., even if medical care is not differentially more productive for high vs. low-risk women, there is still a good reason to target this population to the extent that they account for a disproportionate share of deaths. We hypothesized that in this setting women might have poor information about their risk status — because of a lack of formal schooling, because they do not routinely use health care services, or because of informational constraints more generally — causing them to 'underuse' medical care. By this, we mean that the observed level of utilization is lower than it would be under full information. In line with this conjecture Figure 1 shows that many women have a poor understanding of pregnancy risk factors.

We hypothesized that providing information would help to correct this inefficiency—at-risk women, given a credible signal that they were more likely to experience a bad outcome, would be more likely to seek out health care services. Though we tried to avoid this by urging all women, regardless of their risk status, to seek care it is also possible that the converse might also be true—women given a signal that they were not at risk, might reduce their propensity to use health care services.

3.1. Baseline

The enrollment interview collected information from each woman about her age,12 ethnicity, religion, level of schooling, and reproductive history. We measured her knowledge of pregnancy risk factors—we asked about four pregnancy risk factors (first delivery, older than 35 years, five or more previous deliveries, anemia) and one non-risk factor (intercourse during pregnancy). For each, we asked if it increased, decreased, or neither increased nor decreased the chance of a child being stillborn or dying soon after birth. We also elicited risk beliefs by asking each participant whether she thought she had any pregnancy risk factors (this was a simple yes or no question). We also measured women's expectations about the likelihood of experiencing a stillbirth (in general and for a woman just like her).13 Lastly, we collected some information about each household including number of household members and ownership of various household assets (radio, television set, bicycle, motorcycle, generator, etcetera). These interviews took place between July 2019 and April 2020 (90% of interviews were competed by the end of November 2019). In total we enrolled 36,607 women. For the analysis we exclude 21 women with missing baseline data, leaving us with 36,586 women.

3.2. Follow-up

Return visits to study sites began in November 2019. Study participants who were still pregnant when contacted again completed a short follow-up interview during which we collected information about health care use, food consumption, maternal labor supply, and time use in the month preceding the interview. Conditional on still being pregnant at contact, the probability of selection for the short follow-up interview was 0.33 (later on it was increased to 0.5 to ensure that we had a sufficient number of women). Moving forward we will refer to this as the 'midline' interview.

Study participants who were no longer pregnant completed a longer follow-up interview during which we collected information about health care utilization, the pregnancy outcome and the survival status of the child. This is the endline interview. For women who died prior to the endline interview we interviewed another household member, usually the spouse. Note that all women with a midline interview also later completed an endline interview.

In total we have midline interview data for 5,411 women and endline interview data for 36,152 women. We do not have endline data for 249 participants—177 women could not be located and 72 did not give consent. 185 women also turned out to be ineligible at endline because they were never pregnant. Follow-up interviews took place between November 2019 and December 2020. Incentive payments were made to qualifying women following the endline interview. Verification of utilization and payment was carried out by a separate supervisor. Payments were made in cash. All other participants received small thank-you gifts of nominal value (about $1).

During the endline interview we randomly selected one child under five years old in each participating household from the baseline roster to complete an additional interview. The respondent was usually the mother. This was often, though not always, the enrolled participant. This interview collected information about healthcare utilization by that child and their health status. In total we have data for 21,743 children. We use these data to test for intra-household spillovers.

4.2. Effects on healthcare utilization

The primary outcomes are: (i) any antenatal visit, (ii) four or more antenatal visits, and (iii) a facility delivery. Throughout this paper we use antenatal and prenatal interchangeably. These outcomes are based on participant reports. In the robustness section, we will examine sensitivity to defining these outcomes using documented utilization. In all cases we present results separately for the pre and post-Covid onset samples. The regression results are in Table 6. The effect on the number of visits is in Supplementary Table A.3.

Effects on healthcare utilization

Notes: Estimates are percentage points (i.e., they have been pre-multiplied by 100). This table shows the effect of incentive payments of N2500, N4000, and N10000 (compared to no payment), and information about pregnancy risk, on the probability of (a) any prenatal care; (b) at least four prenatal visits; and (c) delivering in a health facility. The payments were conditioned on the latter two. Pre/post-Covid identifies the sample of mothers with a pregnancy that concluded before/after the start of the Covid-19 pandemic. Estimates are from linear probability models. All models include state, clinic treatment, and year-month of interview dummies. Additional controls are chosen using double-selection lasso. Standard errors in parentheses are clustered at the site level.

$p < 0.1$

$p < 0.05$

$p < 0.01$.

Data source: Endline.

For each outcome we see the same pattern—a large jump in utilization going from N0 (no incentive) to a N2500 incentive and a much flatter response to increasing the size of the incentive. In the short model, facility deliveries increase by 27, 28 and 31 percentage points, respectively (off a baseline of 28

percent). The differences across incentive arms are largely statistically insignificant. The effect of risk information is small and statistically insignificant in nearly all cases. In the long model, for 4+ antenatal visits and a health facility delivery, information appears to have a small negative effect that is counteracted by incentives. The interaction is statistically significant at the 10 percent level for the N10000 payment.

Table 7 examines facility choice. There are a range of healthcare facilities within the formal sector that households can choose from—from health centers to public or private hospitals—that vary in cost and quality. While larger payments might not significantly impact the extensive margin decision, they might impact the type of facility that a woman chooses, conditional on making the decision to use formal health care. We are particularly interested in whether larger payments might induce women to choose higher quality/higher cost facilities. We use hospitals as a rough proxy for these facilities.20 To test this we define a set of indicators for the place of delivery—at home, in a public health center, in a public or private hospital, or other location. As expected, we see a shift from the informal to the formal sector. About 90% of these deliveries are going to health centers. While the results show that women offered a N10000 payment are more likely to choose a hospital delivery, they do not appear disproportionately more likely to do so. Among women given information there seems to be a small but statistically significant substitution in the other direction—away from hospitals towards health centers.

Quality substitution (Effect on place of delivery)

Notes: Estimates are percentage points (i.e., they have been pre-multiplied by 100). This table shows the effect of incentive payments of N2500, N4000, and N10000 (compared to no payment), and information about pregnancy risk, on the choice of delivery facility. Pre/post-Covid identifies the sample of mothers with a pregnancy that concluded before/after the start of the Covid-19 pandemic. Home is a dummy for a home delivery (in the participant's home or the home of someone else). Hospital is a dummy for a hospital delivery (public or private). Health center is a dummy for a health center delivery. Other is a dummy for a delivery in some other location (maternity home, church or spiritual house). Estimates are from linear probability models. For the long model see Table A.4. All models include state, clinic treatment, and year-month of interview dummies. Additional controls are chosen using double-selection lasso. Standard errors in parentheses are clustered at the site level.

$p < 0.1$

$p < 0.05$

$p < 0.01$.

Data source: Endline.

Robustness checks are reported in the Supplementary Appendix. Table A.5 examines whether the results are sensitive to defining the primary outcomes using only documented utilization. To carry out this check we redefine each of the outcomes based on whether the woman was able to provide supporting documentation. If she reported, for example, that she delivered in a health facility but did not have any documentation (14% of women), for the purposes of this analysis she is re-coded as a zero. The source of documentation in most cases is the woman's antenatal card. While this is a potentially more objective record of utilization, women do not always have these cards, even if they used services, because card stock-outs in the health facility are not uncommon. Using documented utilization may therefore bias the results in favor of women in the incentive arm, who are more likely to have these cards (because documentation was required for payment). All the key results carry through.

Table A.6 and A.7 examines sensitivity to the choice of pandemic start date. We show that moving the date forward or back by a few weeks does not impact the results. Tables A.8 and A.9 examine whether the results are sensitive to a woman's timing of enrollment. Recall that sample enrollment across sites took several months. We are less concerned, however, about cross-site interaction, given the distance between sites, and more concerned about within-site interactions (enrollment in a site took several days to complete). In incentives sites there is a concern that women enrolled later in the cycle might be different because they had heard about the cash payment from women enrolled earlier. We find similar results for women enrolled earlier vs. later within a study site. Table A.10 also shows broadly similar results for women in study sites visited earlier vs. later (cross-site interaction).

4.3. Effects on child mortality

Following Okeke and Abubakar (2020), we examine fetal losses (miscarriages), fetal deaths (stillbirths) and newborn deaths. A fetal loss (fetal death) is defined as a pregnancy loss, generally before (after) the 28th week of pregnancy.21 A newborn death is defined as a child death in the first month following delivery. All mortality outcomes are reported per 100 participants. The regression results are in Table 8.

Effects on child mortality

Notes: Estimates are percentage points (i.e., they have been pre-multiplied by 100). This table shows the effect of incentive payments of N2500, N4000, and N10000 (compared to no payment), and information about pregnancy risk, on the probability that a participant experienced: (a) a fetal loss (a pregnancy loss generally before 28 weeks of pregnancy), (b) a fetal death (an in utero child death generally after 28 weeks of pregnancy) and (c) a newborn death (a child death within the first month after birth). Pre/post-Covid identifies the sample of mothers with a pregnancy that concluded before/after the start of the Covid-19 pandemic. Estimates are from linear probability models. All models include state, clinic treatment, and year-month of interview dummies. Additional controls are chosen using double-selection lasso. Standard errors in parentheses are clustered at the site level.

$p < 0.1$

$p < 0.05$

$p < 0.01$.

Data source: Endline.

It is immediately clear from looking at the results that conditional incentives lead to a substantial reduction in in utero child deaths, echoing the findings in Okeke and Abubakar (2020). But how does mortality change in response to larger incentives? Consistent with the healthcare utilization results we find that the effects are concentrated on the extensive margin between no incentive and some incentive. Increasingly larger payments do not buy further measurable reductions in mortality. Fetal losses decrease by 1, 1.1 and 1.2 percentage points, respectively, in the N2500, N4000 and N10000 arms (off a baseline of 3.4 percent) and fetal deaths decrease by 1.2, 1.1 and 1.1 percentage points respectively (off a baseline of 5 percent). Again similar to Okeke and Abubakar (2020) we find no effect of conditional incentives on newborn deaths.22 Across all outcomes we find no measurable effect of risk information.

These mortality reductions largely disappear during the pandemic. This does not appear to be driven by health care utilization. While there are changes in utilization patterns—treated women, for example, were less likely to use hospitals during the pandemic—it seems unlikely that this is responsible for this result. Another possibility that we mentioned earlier is that this might be due to treatment heterogeneity. We showed differences in regional composition in the pre/post-Covid samples. To examine whether this might help explain this result, we re-weight the post-Covid sample to match the pre-Covid sample and run the same regression including these weights. The weighted and unweighted results are nearly identical (see Table A.14) suggesting that this is not responsible. There are two other possibilities: the first is changes in quality. There is evidence that quality significantly deteriorated during the pandemic (see for example Okeke, 2022; Kidanemariam et al., 2025). As such, even if treated women continued to use health care services at similar rates, the returns to these services may have declined during the pandemic. A second possibility is that treated women, by continuing to use health care services, may have faced greater risks of exposure to the virus, exposing them to worse pregnancy outcomes (Balachandren et al., 2022; Gajbhiye et al., 2021). We do not have the data to test these explanations and so leave it to future work.

Additional robustness checks are reported in the Supplementary Appendix. Tables A.15 and A.16 examine sensitivity to the choice of pandemic start date. Tables A.17 and A.18 verify that the results are not driven by a particular state by iteratively dropping one state at a time. Table A.19 examines sensitivity to dropping early stage pregnancies. This is to address the possibility of pregnancy misreporting. A woman might report that she is pregnant, even when she is not, in order to be eligible to participate (in anticipation of getting pregnant later on). The rationale for excluding early stage pregnancies as a test is that misreporting women are more likely to report an early-stage pregnancy as pregnancies at this stage are not obvious. The results survive. Table A.20 subjects the fetal loss result to some additional checks. First, we exclude any women who were past 28 weeks at enrollment. Our findings are the same (the magnitudes are larger). Second, to provide additional reassurance that the results are not an artefact of baseline differences in pregnancy history, we restrict the analysis to women for whom this would be their first delivery. This is a very stringent test as the sample size drops by two-thirds but we see the same pattern of results.

Overall attrition is very low but women in the N4000 and N10000 arms are marginally less likely to drop out (see Table A.21). It is important to put this into proper perspective as the magnitudes are very small, amounting to 20–25 more women remaining in the sample. One does not expect this to make a difference. To examine how this might impact the results we have to make some assumptions about the unobserved outcomes of attriting women. First, we assume that the treatment is completely ineffective for attriters in the treatment group by assigning them the same mortality rate as the control group. Attriters in the control group are assumed to experience similar mortality as their non-attriting counterparts. We then re-estimate the model with these imputed outcomes for attriters. The results are in Tables A.22 and A.23. We see that the results are, in all respects, almost identical to the main results. As an additional test we assume that attriters in the treatment group experience higher mortality than the control group (i.e., they are negatively selected). The results are largely unchanged even if mortality among attriters in the incentive arm were to be as much as five times higher than for the average participant in the control group. Attrition thus cannot plausibly explain these results.

If incentives bring in a lot of new users, this might cause congestion leading to lower quality for the infra-marginal user (Hoe, 2022; Andrew and Vera-Hernández, 2024). This would lead us to underestimate the true mortality effect. There are two reasons why we do not think that this is an important concern in this context. First, health facilities are significantly under-utilized, and second, the number of treated individuals per site is small. Consider that the average health center takes about 19 deliveries per month, i.e., less than one per day. If facility delivery rates in incentive areas increased by 30 percentage points this would translate to about five additional deliveries per month.23 This is very unlikely to lead to congestion. One way to empirically test for congestion-related effects is to exploit cross-site variation in the expected number of participant deliveries per month. This is a function of the number of participants and the (predetermined) month of conception. It varies from 6 to 33 (mean = 13, SD = 3.4). We divide this by the typical number of deliveries per month in the health center and use this as our proxy for congestion. The median value is 0.93, i.e., the median clinic will see a 93 percent increase in monthly deliveries if all study participants give birth in the clinic (values range from 0.09 to 78).24 We assume that the larger the potential increase, the greater the pressure this places on existing capacity. If capacity cannot expand this leads to congestion. Because of the extreme skew we log transform this variable and interact it with the treatments. In general the results do not show evidence of congestion-related effects (see Table A.24).25

Figure 4 examines whether there is heterogeneity in mortality effects by household wealth. Cash transfer programs are almost always aimed at the poor, whether explicitly or implicitly, making it relevant to study benefit incidence (Alatas et al., 2012). Examining heterogeneity by wealth is also relevant because poverty is in itself a risk factor for poor outcomes (Aminu et al., 2014). Do poorer households derive greater benefits from conditional cash incentives? To answer this question, we interact the treatments with indicators for each wealth quartile. We created the wealth index using principal component analysis. To maximize power, and in light of the non-linearity, the incentive treatment is a binary (any incentive vs. no incentive). In general, we do not find strong heterogeneity by wealth. Reductions in fetal deaths are concentrated in the middle of the household wealth distribution but we fail to reject a joint test that all interaction terms are zero.26

Figure 5 tests for heterogeneity by pregnancy risk. Do higher-risk women derive more benefit from medical care usage? As we noted earlier this is a sufficient but not necessary condition. If high-risk women account for disproportionately more deaths this is enough reason to prioritize this group. Indeed, compared to women with no risk factors, newborn mortality is 26 and 81 percent higher, respectively, among women with one risk factor (43% of women) and two or more risk factors (8% of women). Stillbirth rates are 15 and 26 percent higher, respectively. To test whether medical care is more productive for high-risk women, we interact the treatments with a dummy for women with at least one risk factor. None of the interactions are statistically significant.

4.5. Spillovers

Might the effects of in utero incentives persist beyond birth? This is an interesting question with obvious policy implications. It is also one that we can credibly answer in this experiment. Such ‘temporal’ spillovers might pass through several potential channels. The first, one can think of as a ‘conveyor belt’ effect. In health care delivery, services are often linked: for example, a lab test often follows closely on the heels of an office visit, a mother who delivers in a health facility might be more likely to use postnatal care because she is scheduled by default at the time she is discharged. A second channel is through lowering the future costs of using health care services. Individuals might learn more about which services are offered and when. A third channel is through direct changes in health knowledge. Health education, for example, is a typical component of prenatal care visits.

To test for temporal spillovers we look at two outcomes: the first is postnatal care. Postnatal care is universally recommended by the World Health Organization (World Health Organization, 2013). During postnatal visits the health care worker carries out assessments of both mother and child to identify and manage any potential health problems. One might not be very surprised to find effects of incentives conditioned on pregnancy and delivery care “spilling over” to postnatal care because this is generally considered part of the continuum of pregnancy care, triggering a conveyor-belt effect. The second outcome, and arguably a more definitive test of temporal spillovers, is an acute-care (sick child) visit. This is less directly related to parturition care and is temporally more distant.

We have data on whether study participants received any postnatal care and whether surviving index children were sick with a fever, cough or diarrhea in the two weeks preceding the endline interview—and if so whether they were taken to a health facility for treatment. 24% of children in the sample were sick with a

fever, cough or diarrhea in the last two weeks.27 In this setting children are often not taken to a health care facility for treatment when they are sick, contributing to poor outcomes (Adhvaryu and Nyshadham, 2015; Akinyemi et al., 2019). Indeed, less than half of sick children in our sample were taken to a health care facility for treatment. For children not treated in a health facility, the most common place where treatment was received was at a patent medicine vendor.28 In 20–40% of cases, depending on the symptoms, no treatment was sought.

The regression results are in Tables 9 and 10. We do indeed find substantial spillovers. First, mothers offered incentives are about six percentage points more likely to receive some postnatal care (off a baseline of 8 percent).29 Sick index children are also about seven percentage points (about 17 percent) more likely to be treated in a health care facility. We show that this has tangible health consequences. Diarrhea causes nearly 500,000 child deaths every year (more than malaria) and accounts for about nine percent of deaths among children 1–59 months old (Perin et al., 2022). Nearly all of these deaths can be prevented with a cheap and highly effective treatment—oral rehydration salts (Lancet, 1978; Munos et al., 2010). We show that index children with diarrhea are substantially more likely to receive oral rehydration salts.

Spillover effects: Postnatal visits

Notes: Estimates are percentage points (i.e., they have been pre-multiplied by 100). This table shows the effect of incentive payments of N2500, N4000, and N10000 (compared to no payment), and information about pregnancy risk, on postnatal visits. Pre/post-Covid identifies the sample of mothers with a pregnancy that concluded before/after the start of the Covid-19 pandemic. Estimates are from linear probability models. All models include state, clinic treatment, and year-month of interview dummies. Additional controls are chosen using double-selection lasso. Standard errors in parentheses are clustered at the site level.

$p < 0.1$

$p < 0.05$

$p < 0.01$.

Data source: Endline.

Spillover effects: Sick child visits and diarrhea treatment

Notes: Estimates are percentage points (i.e., they have been pre-multiplied by 100). This table shows the effect of incentive payments of N2500, N4000, and N10000 (compared to no payment), and information about pregnancy risk, on the following outcomes: (i) a sick-child visit by the index child, (ii) ORS (oral rehydration salts) treatment if the index children was sick with diarrhea, and (iii) a sick-child visit by another child in the household. The acute care sample includes all index children who were sick with a fever, cough or diarrhea in the two weeks before the endline interview (Data source: Endline). The ORS sample includes all index children in the sample who were sick with diarrhea in the two weeks preceding the endline interview (Data source: Endline). The other child visit sample consists of all children in the other child dataset who were sick with a fever, cough or diarrhea in the two weeks preceding the endline interview (Data source: Other child). Estimates are from linear probability models. All models include state, clinic treatment, and year-month of interview dummies. Additional controls are chosen using double-selection lasso. Standard errors in parentheses are clustered at the site level.

$p < 0.1$

$p < 0.05$

$p < 0.01$.

In Table 10 we also examine whether incentive effects might also spill over to other (non-treated) children in the household. Aside from the channels already discussed, scale/scope economies might also play a role—if a mother is taking one child to see the doctor the marginal cost of taking along another child is low. Recall that we have data for other young children in participating households. We know whether the child was sick in the preceding two weeks and whether they were taken to a health care facility for treatment. We find evidence of intra-household spillovers—acute care visits also increase for other children in the household. The increase is large and statistically significant for children in the N4000 and N10000 arms. Further analysis in Table A.28 suggests that this might be related to scale/scope effects. The effects are coming primarily from households where the index child was also sick.

5.1. Why does child mortality decrease?

Okeke and Abubakar (2020) also explored this but we have the advantage of richer data that provide more definitive evidence. The most straightforward explanation is that mortality decreased due to increased use of medical care by mothers induced by the cash incentives. Prenatal care, for example, is believed to be important for fetal/infant outcomes. During prenatal visits health providers can detect (and treat) problems that, if left untreated, would cause poor outcomes. Regular blood pressure checks help to detect maternal hypertension in pregnancy, a complication that leads to poor fetal outcomes (Ahmad and Samuelsen, 2012; Lindheimer et al., 2008). Malaria and anemia prevention/treatment during pregnancy also leads to better fetal outcomes (Stephansson et al., 2000; Dorman and Shulman, 2000; Van Geertruyden et al., 2004). Anemia in pregnancy is a particularly pernicious problem in sub-Saharan Africa—a setting where many mothers are malnourished—where it is estimated to affect up to half of all pregnant women (World Health Organization, 2021).

Consistent with this, Table 11 shows that mothers offered incentives are 12 to 14 percentage points more likely to receive a blood pressure check during the pregnancy, 13 to 14 percentage points more likely to carry out a laboratory test, and 10 to 11 percentage points more likely to take iron supplements and malaria prophylaxis. These results may understate the impact because they only measure whether mothers ever received these interventions and not how frequently they did. The frequency may be even more critical—malaria prevention, for example, depends on regularly taking prophylactic medication. Further showing why this is important, while 76 percent of women in the control arm attended antenatal care, about one-quarter did so only once or twice. To the extent that mothers in the incentive arms attended more prenatal visits, they would have been not only more likely to receive these interventions but to receive them more consistently over a longer period.30

Medical care as a pathway

Notes: Estimates are percentage points (i.e., they have been pre-multiplied by 100). This table shows the effect of incentive payments of N2500, N4000, and N10000 (compared to no payment), and information about pregnancy risk, on the outcomes shown in the table header. BP denotes a blood pressure check, Lab denotes a laboratory test, Anemia denotes whether the mother took iron/folic acid supplements, and Malaria denotes whether the mother took malaria prophylaxis, all during the pregnancy. Doctor identifies whether care was provided by a doctor during the pregnancy or delivery. Pre/post-Covid identifies the sample of mothers with a pregnancy that concluded before/after the start of the Covid-19 pandemic. All models include state, clinic treatment, and year-month of interview dummies. Additional controls are chosen using double-selection lasso. Standard errors in parentheses are clustered at the site level.

$p < 0.1$

7. Conclusion

This paper has presented results from a randomized experiment in which expectant mothers in Nigeria were offered conditional cash payments to use formal health care services (prenatal and delivery care in a health care facility). Participating mothers were randomized to one of three payments: N2500, N4000, or N10000. Some mothers were also given an information signal about their pregnancy risk—they were notified if they possessed certain characteristics associated with a higher probability of an early child death.

Our key finding is that the incentive response function is highly non-linear. The effects are concentrated on the margin between no incentive and some incentive, with rapidly diminishing returns to incentive size. Child mortality effects are indistinguishable across payment levels. A key policy implication is that even very large payments may not achieve desired utilization goals. Extrapolating from our data suggests that it might take payments an order of magnitude larger than the amounts offered in this experiment to get to 80 percent utilization (a common target). Combining smaller incentives with interventions targeting non-cost barriers may be more cost-effective.34 Eliminating non-cost barriers would have the effect of shifting the entire demand curve up, whereas sliding along the curve by offering more money would generate only modest increments in utilization. We speculate that the discontinuous change in uptake between N0 and N2500 may be due to the zero-price effect (Shampanier et al., 2007; Iizuka and Shigeoka, 2022) as even the smallest payment fully covered the cost of care for most households.

On the whole we find largely null effects of providing information about pregnancy risk. We feel confident in stating that it does not lead to more healthcare utilization. We find that high-risk women given information about their risk status used medical care at similar rates as their counterparts in the control group. We provide evidence that the reason may be that the new information did not substantially change women's beliefs about their risk.

We conclude with the recommendation that conditional incentive programs targeting in utero use of health care services should be expanded in developing countries. This recommendation is stronger in the light of new evidence we have uncovered that it generates quantitatively and qualitatively important spillovers. By showing how incentive effects vary with incentive size, we provide policymakers with critical new insights. Additionally, we have presented new evidence of how demand-side incentives interact with the supply-side. The key lesson is that incentives can reduce high rates of child mortality even under status quo conditions, but investing in improving service delivery at the same time is likely to generate even larger payoffs.

Does it pay firms to register for taxes? The impact of formality on firm profitability

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1. Introduction

Informality is pervasive amongst firms in developing countries, yet simple comparisons of formal and informal firms usually reveal that formal firms are more productive and profitable. A series of high-profile sector studies by the McKinsey Global Institute around the world comparing the operation of formal and informal firms concluded that informality has a very negative impact on productivity, even going so far as to conclude that "in Portugal and Turkey, for instance, informality accounts for nearly 50 percent of the overall productivity gap with the United States" (Farrell 2004). However, such estimates ignore the fact that formality is a choice of firms – the lower productivity of informal firms may therefore just reflect less productive firms choosing to remain informal rather than be the consequence of informality.

This paper seeks to provide more credible evidence of the impact of being formal on business profits, using new survey data on firms in urban Bolivia, the country with the highest levels of informality in Latin America (World Bank, 2007). We control for a rich set of owner characteristics, which provide a more comprehensive set of measures of owner ability, background, and motivations for entering business than existing studies of formality. We then identify the impact of registering for a tax identification number on firm profits by using the GPS-measured distance of a firm from the tax office where registration for taxes occurs as an instrument for whether or not a firm is registered for taxes. After controlling for the distance of the firm to the city center and other locational characteristics, we argue that this distance affects the information a firm has about registration, but does not independently affect profits. We show our results are robust to several potential threats to this identification assumption.

We find that registering to pay taxes leads to significantly higher profits for the firms that the instrument affects. Nevertheless, we also provide evidence of heterogeneous effects of formality on profitability. In particular, we find that although registering for taxes

increases profits for firms in the middle size group in our sample (2 to 5 workers and the middle tercile of capital stock), registering for taxes is associated with lower profits for firms smaller than this, and for firms larger than this. The main benefit of registering for taxes appears to be an increase in the customer base through the ability to issue tax receipts – we find no evidence of increased access to finance. Very small firms are too small to benefit from issuing tax receipts, while owners of large informal firms have high ability and can achieve a large customer base through their own business skills.

This research adds to a nascent literature on the micro-level impacts of informality on firms. Despite increasing research issue concerning the nature of informality, its determinants, and its macro effects (e.g. Loayza (1996), Schneider and Enste (2000), Maloney (2004, 2006), Perry et al. (2007)), there are currently few studies which attempt to provide rigorous estimates of the impact of formality on firms themselves. In a recent study, Fajnzylber, Maloney, and Rojas (2006a) use propensity-score matching methods and control function approaches to estimate the impact of paying taxes and belonging to business associations on the performance of Mexican micro enterprises. They find relatively large impacts, with paying taxes estimated to increase business profits by at least 20 percent, and belonging to business associations estimated to increase business profits by at least 10 percent. However, their identification relies on assumptions about the formality status of firms being determined either on the basis of a set of observable variables or through a specific functional form in the estimation equation. If firms select into formality on the basis of unobserved owner ability or firm productivity, these may overestimate the impact of becoming formal.

A second set of recent studies have looked at the short-term impacts of increases in formality induced by business simplification procedures. Monteiro and Assunção (2006) and Fajnzylber, Maloney, and Rojas (2006b) use difference-in-differences and regression discontinuity designs respectively to obtain non-experimental estimates of the impact of the SIMPLES program in Brazil. They find higher revenues for firms that operate with a license, but do not examine whether these revenues are offset by higher expenses (such as taxes), or whether these higher revenues are also accompanied by higher profits. Two

2.1 Sampling Design

Six industries were chosen for the survey: grocery stores, restaurants and food sales, manufacturing of clothing from wool and cloth, transportation of passengers and cargo, manufacturing of clothing from camelid wool (from llamas and alpacas), and manufacturing of furniture from wood. The industries were chosen to represent a large portion of the self-employed and small employers, and to encompass a diversity of sectors. According to the 2005 MECOVI (Bolivian Living Standards Measurement Survey), the industries chosen include four of the top five industries[3] for urban small and medium enterprises and cover approximately 40 percent of all self-employed and employers.

The sample frame consisted of a geographic information database maintained by the survey firm. This database is based on a census of all economic establishments in these cities carried out in August 2005, and includes enterprises operating within households. This was supplemented with data for the transportation sector on all firms that have registered their cars. This provided a reasonably comprehensive sampling frame for urban areas.[4] The sample was stratified across cities and firm size, in order to include a mix of micro firms (with less than five workers), and small firms with 5-20 workers. The sample used here is almost equally divided across the four cities. Groceries, Food, and Transportation each constitute about 20 percent of the sample, Clothing from wool and cloth and manufacturing from wood each constitute 15 percent of the sample, and clothing from camelids the remaining 10 percent.

2.2 Sample characteristics

Table 1 provides summary statistics for the 469 firms used in this paper. The median firm has been in business for 9 years, although 27 percent have been in business 3 years or

[3] The only one of the top five industries not covered is construction, where it was felt there was little overlap between formal and informal firms, and where the broad industry grouping is very heterogeneous, covering skilled engineers and architects along with less skilled workers.

[4] A sample frame was not available for rural areas, and therefore snowball sampling methods were used to survey camelid and wood firms in rural areas. The lack of a representative sampling frame provides a further reason for excluding rural areas in this study.

less. The median firm in the sample has 2 paid workers, with only 20 percent of the sample having five or more paid workers. Mean monthly profits are 1628 Bolivianos ($US211).[5] Profits were measured through a direct question, following the recommendations made in de Mel, McKenzie and Woodruff (2007). Half the firm owners are female. Average education levels for owners are quite high, but there is substantial variation. The median owner has 12 years of education, but 20 percent of firm owners have less than 6 years education, and 31 percent have more than 12 years.

2.3. Measuring Owner Ability and Background

A main concern for our empirical work is that there are characteristics of firm owners which affect both profitability and the decision of whether or not to formally register the business. One such factor could be owner ability, which may determine the size of the gain in profits from becoming formal. Other such factors could be family background and wealth, which might affect the ability of the firm to meet the costs of formalizing, and the utility associated with being formal. Our instrumental variables strategy aims to overcome such concerns, but we also attempt to directly measure and control for these variables much more than is possible with variables such as gender, own education, age, marital status and ethnicity found in standard firm surveys.

We use three different measures of owner ability and motivation. The first is mother's education, consisting of dummy variables for mother having no education (36 percent of firm owners), and for mother having 9 or more years education (16 percent). The second measure is entrepreneurial self-efficacy. Self-efficacy is the first principal component of ten questions intended to measure the self-assessed ability to perform certain tasks. Owners were asked on a four point scale how confident they are that they could do ten business tasks. Examples include their ability to estimate accurately the costs of a new project; to resolve a difficult dispute with a client or supplier in another city; to hire good employees to expand their business; to sell a product to a new client; and to price their business correctly if they wish to sell it. Such a measure is more closely tied to business skills than years of education, and has a correlation of only 0.11 with years of education.

[5] During March 2007, 1 USD was approximately 7.7 Bolivianos.

either be formal or everyone informal. If there are heterogeneous effects, those who are informal will not be appropriate controls for those who are formal.

However, equation (1) also shows that some firms for which it is profitable to become formal will not do so if the initial costs of registering are too high. In the Bolivian case, the monetary costs of registering for a NIT are very low (zero for the registration itself, with only the costs of photocopying and obtaining accompanying documents needed). However, the time and information costs will depend on how closely located the firm is to the office where they have to register, and on how much general information is available about the registration process. This provides a potential source of identification: comparing two firms with the same potential net change in profits (and same psychic benefits) from obtaining a tax identification number, but with different time and information costs will enable estimation of the increase in profits resulting from formalization.

4.2 Identification

Our basic identification strategy is therefore to use the geographic location of the firm relative to that of the tax office as a source of variation on the information and time costs of registering, assuming that after other locational controls are added, that distance to the tax office has no independent impact on profits.

The latitude and longitude of each firm was obtained using GPS receivers. In each city there is a single location of the tax office where firms can register for a NIT. We also obtained the coordinates of these tax offices, and of the city centers. We then calculate the straight-line distance from the firm to the nearest tax office and to the city center. Since the sample is contained in dense urban neighborhoods, straight-line distances will be good approximations for actual travel distances (Gibson and McKenzie, 2007). We have restricted the sample to firms within a 10 kilometer radius of the tax office: the mean firm is 3.2 kilometers away. The assumption is that for two firms of equal distance from the city center, the one that is closer to the tax office has better knowledge of the registration process.

Our survey indeed reveals limited knowledge about the process of getting a tax number. 39 percent of firms without a NIT say they don't know what a tax number is. Only 44 percent of unregistered firms say they know where the nearest tax office is, and only 31 percent are able to give its location to within 3 kilometers. Table 2 uses the sample of unregistered firms to examine whether there is a relationship between distance to the tax office and knowledge of where the tax office is. Column 2 shows a large, but not significant effect on whether the firm owner says they know where the office is. Since some firm owners may claim to know where it is, but not actually know, we also asked them to tell us the distance to the office. Columns 3 and 4 show that there is a large and significant effect of distance to the tax office on this measure of ignorance.

Another reason that distance to the tax office could determine whether or not a firm registers is that the chance of receiving a visit from a tax inspector may be greater the closer a firm is to the tax office. Column 7 of Table 2 shows that this is indeed the case when we don't condition on firm and location characteristics. However, in all of our regressions we will include controls for city, industry, the average tax enforcement rate in each city*industry pair, and the average tax enforcement rate in a 1 kilometer radius around the firm.[6] Column 8 shows that after including these controls, there is no effect of distance to the tax office on enforcement.

We thus believe that after the inclusion of firm, owner, and locational controls, distance to the tax office is determining the information a firm has about registration, but is not having an independent effect on its profits. We examine possible threats to this exclusion restriction in the robustness section below.

4.3. First-stage: does distance to the tax office predict formality?

Table 3 then examines whether log distance to the tax office predicts whether or not a firm has a tax identification number, after controlling for firm owner, firm, and location

[6] In both cases we do not include the firm's own tax enforcement in calculating the average of firms around it.

characteristics. We show marginal results from estimation of a probit equation where having a NIT is the dependent variable. We present two sets of results. Column 1 does not control for firm size, and shows a strong and significant effect of log distance to the tax office on having a NIT. The marginal effect shows that being 3.2 kilometers away from the tax office (the mean) is associated with a 13 percentage point reduction in the likelihood of having a NIT. Since only 29 percent of firms have a NIT, this is a sizeable effect. Column 2 shows this result also holds after conditioning on firm size, as measured by log of capital stock, and dummy variables for the number of paid workers being in different categories. As one would expect, larger firms are more likely to be formal.

Column 1 also highlights the role of owner ability in the formality decision. Owners with more education, higher entrepreneurial self-efficacy, and who went into self-employment for business growth reasons are more likely to have a NIT. After controlling for owner ability, there is no significant effect of gender on the formality decision. Secondly, column 1 shows that firms are more likely to have a NIT when enforcement is higher: both the inspection rate within their city*industry pair and within a 1 kilometer radius of the firm are positively and significantly associated with having a NIT.

In columns 3 and 4 of Table 3, we then examine whether a similar strategy can be used to estimate the impact of a municipal license. Municipal licenses are obtained at the municipal office (*Alcaldia).* However, while there is a weak negative correlation between log distance to the municipal office and having a municipal license (-0.10), this correlation disappears after controlling for firm, owner, and locational characteristics. This is consistent with the view that firms have reasonable knowledge about how and where to get a municipal license, so that distance does not provide sufficient variation in information access to enable identification in this case.

5. The impact of registering for taxes on profits

We now turn to estimation of the impact of a NIT on firm profits. We focus on profits as the dependent variable for two reasons. First, from the point of view of a firm deciding on whether or not to register, the impact on profits is a key determinant. Secondly, whilst from a social welfare point of view one would also like to know whether formality improves productivity, estimation of firm productivity is much more complicated than profitability, and relies heavily on assumptions that are unlikely to be credible with the types of firms surveyed here (see the critique in Katayama, Lu, and Tybout, 2006).

The basic estimation equation of interest is, for firm *i*:

$$\ln(profits)_i = \alpha + \beta NIT_i + \delta' X_i + \gamma' Z_i + \theta' L_i + \varepsilon_i \qquad (2)$$

where NIT_i is a dummy variable taking value one if firm *i* has a tax identification number, X_i, Z_i, and L_i are owner, firm, and locational characteristics respectively. We instrument the tax identification number using log distance to the tax office. We employ two methods of estimating (2). The first is two-stage least squares, which has the advantage of not imposing distributional assumptions on the error term. The second method is maximum-likelihood estimation (MLE) of a treatment effects model, which takes account of the fact that *NIT* is a binary variable, and jointly estimates a probit equation for *NIT* along with equation (2).[7] In both cases we estimate the equation with and without controls for firm size.

Columns 1 and 2 of Table 4 provide the MLE estimates and columns 3 and 4 provide the 2SLS estimates. The estimates show very large, and highly significant, increases in profits from registering for a tax identification number. The dependent variable is log profits, so the MLE estimate in column 2 suggests that firms which obtain a tax identification number have 86 percent higher profits. The 2SLS estimates are larger, but not significantly so. The MLE estimates are significant at the 1 percent level, and the 2SLS estimates at the 5 percent level. Thus according to these estimates, firms earn much larger profits after registering.

[7] This was carried out using the treatreg command in STATA.

Several other variables are also notable in Table 4. First, after conditioning on size and other characteristics, male and female owners don't have significantly different profitability. Secondly, there is a strong and significant effect of entrepreneurial self-efficacy on profits – more able owners earn higher profits. Third, profits are lower when the tax inspection rate is higher, which is consistent with higher taxes reducing profits. Fourth, profits are higher for larger firms. Finally, note the point estimate on log capital stock (which excludes land and buildings), suggests a 10 to 12 percent per month return on capital for the mean firm in the sample. This return to capital is comparable to that obtained by McKenzie and Woodruff (2006) for microenterprises in Mexico.

5.1. Robustness

In Table 5 we examine the robustness of our results to potential challenges to the identification strategy. Column 1 of Table 5 repeats the MLE estimate from column 2 of Table 4: an 86 percent increase in profits from having a NIT. The first potential threat to our exclusion restriction is that more able entrepreneurs may choose to locate their businesses in locations which are closer to the tax office. The fact that so few unregistered firms know where the tax office is also leads us to believe it extremely unlikely that firms are choosing their business location to be closer to or further from the tax office. If they are, unobserved owner ability will not be orthogonal to distance.

Assuming that observed and unobserved ability have similar correlations with distance, in column 2, we examine the sensitivity of our results to dropping our measures of ability and owner background. The point estimate drops from 0.86 to 0.81, with the difference not significant. Therefore the inclusion of ability measures does not significantly affect the estimated treatment effect, which provides support for the view that more able entrepreneurs are not choosing locations closer to the tax office. Instead, micro and small entrepreneurs typically operate at, or near, their homes, making the location decision of the firm a function of where the owner happens to live. In our sample, 33 percent of firms

are operating out of the dwelling of the owner.[8] The point estimate of the treatment effect is still 0.78 for this sub-sample, although with the smaller sample size, it is not statistically significant.

A second potential threat to the exclusion restriction is that despite controlling for the distance to the city center, distance to the tax office may be correlated with other interactions with the government, which could also affect profitability. To examine the robustness of our results to this possibility, in column 3 we add the distance to the municipal office when municipal licenses are issued as a further control. The mean distance from the municipal office to the tax office is only 1.4 kilometers. We see this variable has no effect on firm profitability, and leads to no significant change in the estimated effect of a NIT. This adds further support to our claim that the distance to the tax office is picking up the specific information and time costs of obtaining a NIT, and not something else to do with other interactions with the government.

Finally, while we have controlled for the general tendency of firms in a particular location to receive visits from a tax inspector, in column 4 of Table 5 we also include whether a firm itself has received a visit from a tax inspector. This is likely to be endogenous to profits, but conditioning on this provides a further check of the robustness of our results to controls for enforcement. Again there is no significant change in the coefficient after adding this control, and if anything, the point estimates show even larger impact of a tax identification number. Thus our results appear robust to several potential threats to our identifying assumption.

5.2. Comparison with OLS

Columns 5 through 8 of Table 4 provide OLS estimates of the impact of having a NIT on log profits, for comparison with the treatment effect estimates provided by MLE and 2SLS. We begin by including the same control variables as in columns 1 through 4. Columns 5 and 6 show that OLS provides much lower estimates of the impact of a NIT.

[8] Firm owners were surveyed at their business locations, and therefore we do not have the GPS coordinates of the home locations which would allow us to say precisely how many firms are operating within a kilometer or two of their homes.

Without controlling for firm size, having a NIT is associated with 43 percent higher profits. However, after controlling for size, there is no significant effect of a NIT, with the point estimate only being 1.2 percent higher profits.

However, the typical concern in estimating equation (2) by OLS is that there are unobservable firm or owner characteristics which are correlated with the decision to become formal and which also affect profits. Two of the most common such factors would be owner ability and anticipated positive productivity shocks. In both cases, this would lead to a positive bias in the OLS estimate of the impact of having a NIT. That is, after using instrumental variables, we would expect to find *lower*, not higher, impacts of NIT.

To examine whether the direction of bias in the OLS estimates is working the way we would expect, columns 7 and 8 of Table 4 remove our measures of owner ability, background, and reasons for going into business. If wealthier, more able owners, who expect their businesses to grow are the ones more likely to formalize, then removing these variables should lead to larger coefficients on NIT. This is indeed what we see. When we don't control for firm size and we fail to control for these owner characteristics, the impact of a NIT is to increase profits by 53 percent, compared to 43 percent when these controls are included. While our proxies for ability are more comprehensive than most studies, there may still be unobserved ability components that influence both profits and formality status. However, based on this analysis, we would expect such terms to operate in the same way, if anything lowering the OLS estimates further with their inclusion. Thus it does not appear that this simple story of ability bias can explain the difference between the OLS and MLE/2SLS estimates.

5.3 Treatment effect heterogeneity and the difference from OLS

OLS will also provide a different estimate than the MLE/2SLS estimates if there are heterogeneous treatment effects. That is, if the impact of obtaining a tax identification number differs across firms. If selection into formality status occurs only in terms of the observable variables included in equation (2), and columns 5 and 6 of Table 4, OLS will

estimate the average treatment effect. In contrast, 2SLS and MLE will estimate local average treatment effects (Imbens and Angrist, 1994). Specifically, they will estimate the impact of registering for taxes on firms for which distance to the tax office influences their formality status. These are firms for which distance makes a difference to whether they know how to register, or which are enough at the margin of registering that a small change in the travel costs of registering will influence their registration decision.

There are good reasons to believe that the reasons for being informal vary with firm size, since many of the costs and benefits of becoming formal will vary with firm size. To explore the possibility of different effects of a NIT, we use propensity-score matching to estimate the impact of having a NIT for different firm size groupings.[9] The same set of variables as were used in the OLS regressions are used for matching purposes. Nearest neighbor matching was used to estimate average treatment effects matching firms to their three nearest neighbors[10].

Table 6 reports the results of propensity score matching. The first two columns report the results of matching, with and without controls for firm size. After controlling for firm size, the average treatment effect of a NIT is a 20.9 percent increase in profits – larger than the OLS coefficient, but smaller than the MLE/2SLS coefficients. Again, with heterogeneous treatment effects the propensity-score matching treatment effect will differ from the average treatment effect estimated by OLS and the local average treatment effect estimated by MLE and 2SLS. We then split firms into three size groupings, based on terciles of capital stock and groupings of 0 to 1, 2 to 5, and 6 and above workers. Although the smaller sample sizes when we split the sample this way lead to some statistically insignificant results, the point estimates are suggestive of heterogeneity of treatment effects. Having a NIT is estimated to *lower* profits for very small firms by 61.6

[9] One could also estimate the treatment effects regressions for subsamples. However, with our rich set of controls and smaller samples, convergence of the MLE was a problem for these restricted samples. The propensity-score matching estimators enable the large number of covariates to be employed more efficiently, and thus we use this method to investigate heterogeneity of effects.

[10] Matching was carried out using the nnmatch command in STATA. See Abadie, Drukker, Herr and Imbens (2004).

percent, increase profits for slightly larger firms by 40.8 percent, but then *lower* profits for firms with 6 or more workers in the top tercile of capital stock, by 54.6 percent.

What explains this possible nonlinear relationship between formality and profitability? The explanation for very small firms is clear. These firms are too small to immediately benefit from formalization. Registering for taxes immediately involves more costs in the form of tax payments, but these firms are too small to benefit from increased customer base or better access to credit. However, if these firms plan on growing over time, the current cost of formalizing may be justified in terms of the anticipated future benefits. Consistent with this, we find the firm owners who are formal at this small size to be of much higher ability. Firms who are formal in the middle size group (2 to 5 workers and the middle tercile of capital stock) are big enough to enjoy some of the benefits of formalizing, and tax registration increases profits for this group. The mid-sized formal firm owners are still of higher ability than those who remain informal.

A new insight from this work concerns the somewhat larger firms – those with 6 or more workers and in the top tercile of capital stock. The propensity score matching estimate shows a large and statistically significant *negative* effect of tax formality on profits for this group of firms. The firm owners who have managed to get to this size and remain informal have higher entrepreneurial self-efficacy than firm owners of this size with registered firms. These firms have likely figured out ways to avoid inspections and access many of the potential benefits of formality, without having to pay taxes. Only 25 percent of these larger informal firms received a visit by tax inspectors in 2006, compared to 77 percent of formal firms of the same size.

Our finding that the owners of larger informal firms have higher entrepreneurial ability than owners of larger formal firms is in contrast to the prediction of Rauch (1991), that smaller and informal businesses are more likely to be run by less-talented entrepreneurs. In his model, enforcement only occurs for larger firms, and more talented entrepreneurs are endogenously allocated to bigger firms, where they are compensated for the costs of regulation. However, in practice, the talent of an entrepreneur may also affect his or her

formalizing. As a result, we can estimate the impact of registering for taxes on firms for whom the choice to formalize is affected by their distance to the tax office. We find large effects of registering for taxes on the profits of these firms, with the effects appearing to come mainly through increases in their customer base.

However, we also find evidence that the effect of formalizing differs across firms. While there is a group of firms which can benefit from formalizing, formality lowers profits for very small firms – who are too small to benefit - and for the larger firms in our sample. Firm owners with enough ability to grow their firm to a size of six workers or more and achieve a reasonably large capital stock would have lower profits from formalizing – they would pay more taxes, but are already able to reach a large customer base without formalizing.

The irony then is that registering for taxes would seem to benefit most the informal firms who don't know how to formalize. Conceptually, those who are informal by choice will expect lower profits from formalizing otherwise they would be formal, whereas those who are informal due to ignorance of the procedures needed to formalize stand to benefit from learning how to. Consequently, the results of this study highlight the need for better information provision by Governments to induce firms at the margin to formalize. They also suggest the need for policies which target the high ability owners of larger informal firms, through providing more benefits of formality and tighter enforcement.